# Rapporto Analisi sollecitazione

| | |
|---|---|
| File analizzato: | First arm.ipt |
| Versione di Autodesk Inventor: | 2026.1 (Build 301233020, 233B) |
| Data di creazione: | 23/09/2026, 14:03 |
| Autore studio: | Nuzzolese Alessio |
| Riepilogo: | |

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## Analisi statica:1

**Obiettivo generale e impostazioni:**

| | |
|---|---|
| Obiettivo di progetto | Punto singolo |
| Tipo studio | Analisi statica |
| Data ultima modifica | 23/09/2026, 14:02 |
| Stato modello | [Primario] |
| Rileva ed elimina le modalità corpo rigido | No |

### iProperties

#### Riepilogo

| | |
|---|---|
| Autore | nuzzo |

#### Progetto

| | |
|---|---|
| Numero parte | First arm |
| Progettista | nuzzo |
| Costo preventivato | 0,00 € |
| Data creazione | 23/09/2026 |

#### Stato

| | |
|---|---|
| Stato progetto | In corso |

#### Fisiche

| | |
|---|---|
| Materiale | Generico |
| Densità | 1 g/cm^3 |
| Massa | 0,035762 kg |
| Area | 187,219 cm^2 |
| Volume | 35,762 cm^3 |
| Baricentro | x=4,90638 cm<br>y=1,53691 cm<br>z=0 cm |

Nota: i valori fisici potrebbero essere diversi da quelli utilizzati per FEA riportati di seguito.

**Impostazioni mesh:**

| | |
|---|---|
| Dimensione media elemento (frazione del diametro modello) | 0.1 |
| Dimensione minima elemento (frazione della dimensione media) | 0.2 |
| Fattore di gradazione | 1.5 |
| Angolo di rotazione massimo | 60 gr |
| Crea elementi mesh curvi | Sì |

## Materiale/i

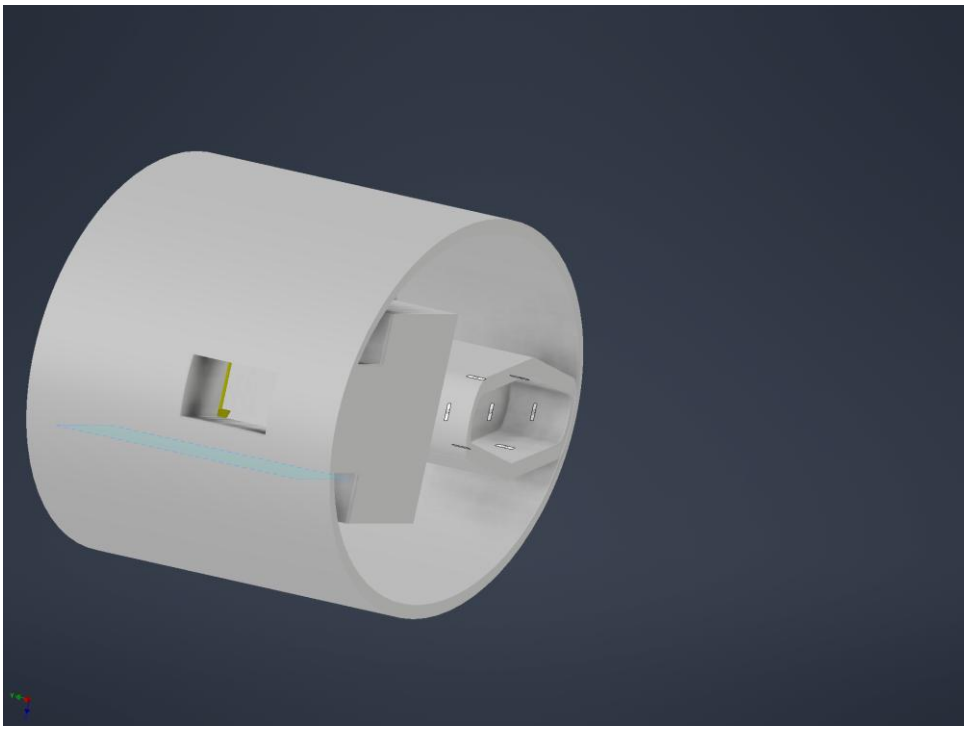
| | | |
|---|---|---|
| Nome | Plastica ABS | |
| Generale | Densità della massa | 1,06 g/cm^3 |
| | Resistenza allo snervamento | 20 MPa |
| | Resistenza massima a trazione | 29,6 MPa |
| Sollecitazione | Modulo di Young | 2,24 GPa |
| | Coefficiente di Poisson | 0,38 su |
| | Modulo a taglio | 0,811594 GPa |
| Nome/i parte | First arm.ipt | |

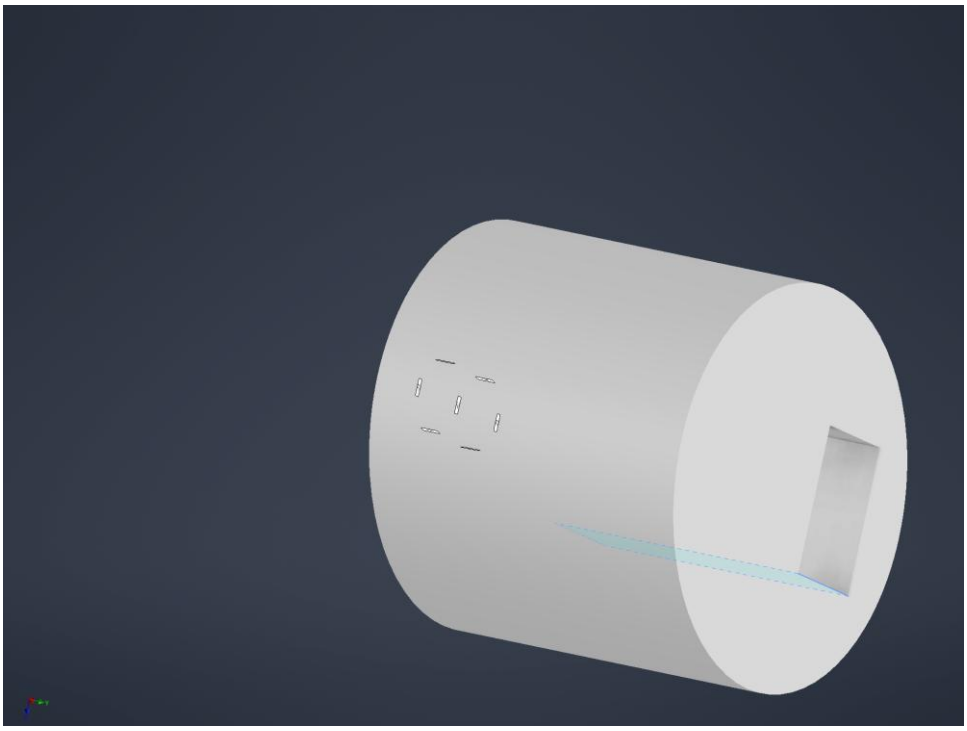
## Condizioni operative

### Forza:1

| | |
|---|---|
| Tipo carico | Forza |
| Intensità | 10,000 N |
| X vettore | 0,000 N |
| Y vettore | 0,000 N |
| Z vettore | 10,000 N |

**Faccia/e selezionate**

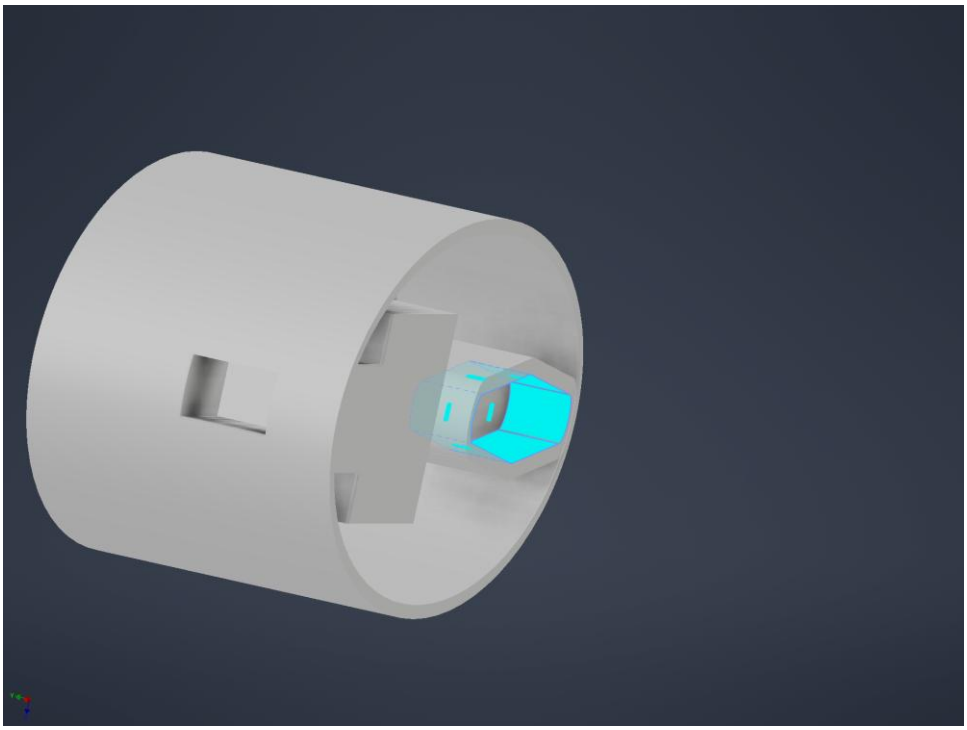


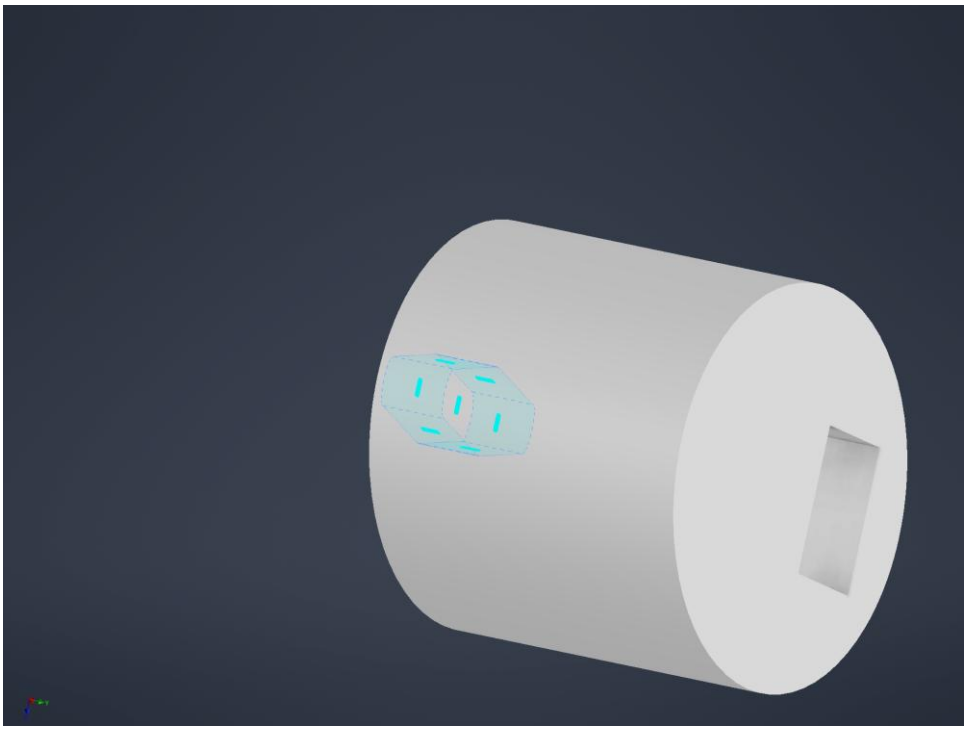


## Vincolo di fissaggio:1

| Tipo vincolo | Vincolo di fissaggio |
|---|---|

**Faccia/e selezionate**





## Risultati

### Forza di reazione e momento su vincoli

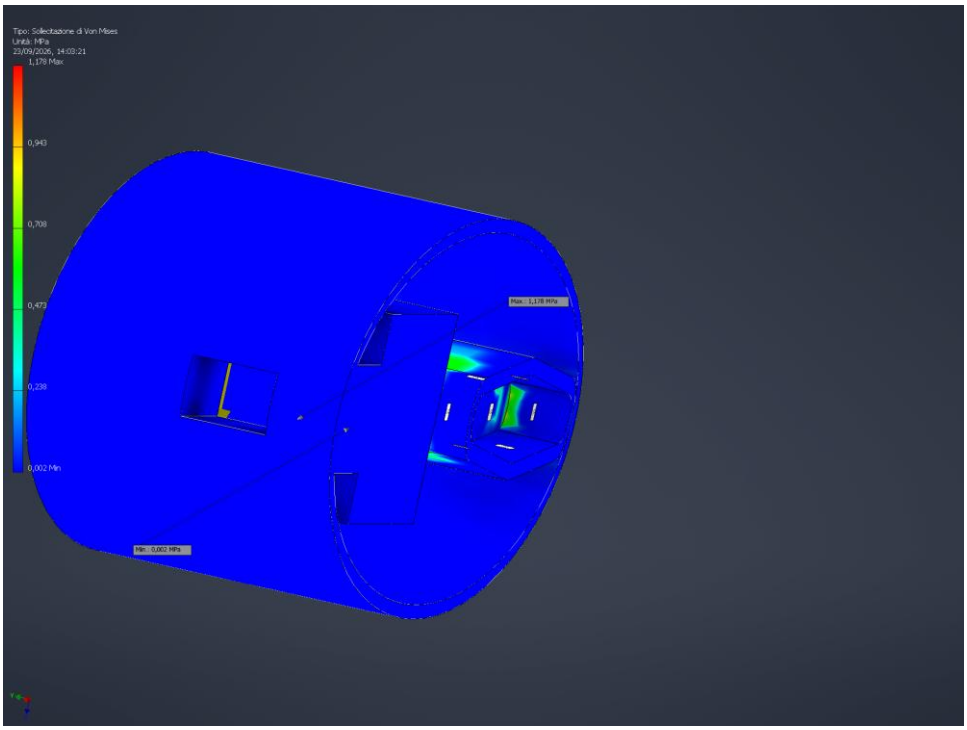
| Nome vincolo | Forza di reazione | | Momento di reazione | |
|---|---|---|---|---|
| | Intensità | Componente (X,Y,Z) | Intensità | Componente (X,Y,Z) |
| Vincolo di fissaggio:1 | 10 N | 0 N | 0,510477 N m | -0,12499 N m |
| | | 0 N | | 0,494939 N m |
| | | -10 N | | 0 N m |

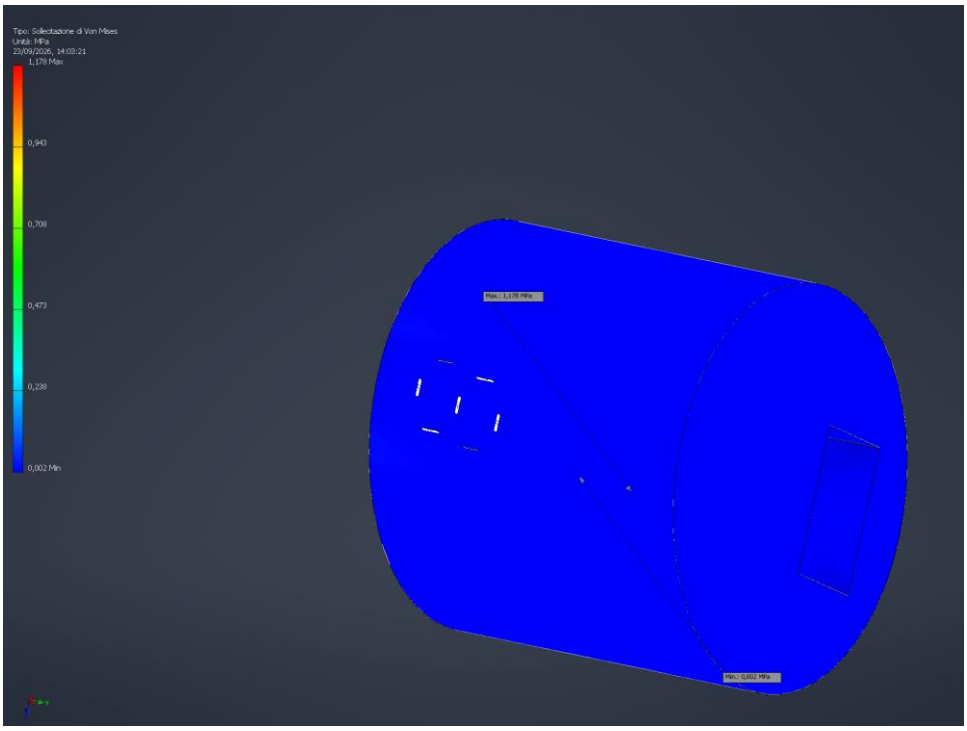
## Riepilogo risultati

| Nome | Minimo | Massimo |
|---|---|---|
| Volume | 35762 mm^3 | |
| Massa | 0,0379077 kg | |
| Sollecitazione di Von Mises | 0,0024196 MPa | 1,17811 MPa |
| Prima sollecitazione principale | -0,537456 MPa | 1,67573 MPa |
| Terza sollecitazione principale | -1,71068 MPa | 0,545762 MPa |
| Spostamento | 0 mm | 0,0441144 mm |
| Fattore di sicurezza | 15 su | 15 su |
| Sollecitazione XX | -0,629358 MPa | 0,626109 MPa |
| Sollecitazione XY | -0,291993 MPa | 0,309292 MPa |
| Sollecitazione XZ | -0,239632 MPa | 0,450889 MPa |
| Sollecitazione YY | -1,39404 MPa | 1,37471 MPa |
| Sollecitazione YZ | -0,488759 MPa | 0,526793 MPa |
| Sollecitazione ZZ | -1,03995 MPa | 1,04685 MPa |
| Spostamento X | -0,00891762 mm | 0,0085901 mm |
| Spostamento Y | -0,00333314 mm | 0,00325105 mm |
| Spostamento Z | -0,00610884 mm | 0,0441139 mm |
| Deformazione equivalente | 0,00000100216 su | 0,000502512 su |
| Prima deformazione principale | -0,00000100664 su | 0,000576889 su |
| Terza deformazione principale | -0,000565571 su | 0,000000518482 su |
| Deformazione XX | -0,000150335 su | 0,000126631 su |
| Deformazione XY | -0,000179889 su | 0,000190546 su |
| Deformazione XZ | -0,00014763 su | 0,00027778 su |
| Deformazione YY | -0,000370502 su | 0,000391436 su |
| Deformazione YZ | -0,000301111 su | 0,000324542 su |
| Deformazione ZZ | -0,000251421 su | 0,000412334 su |

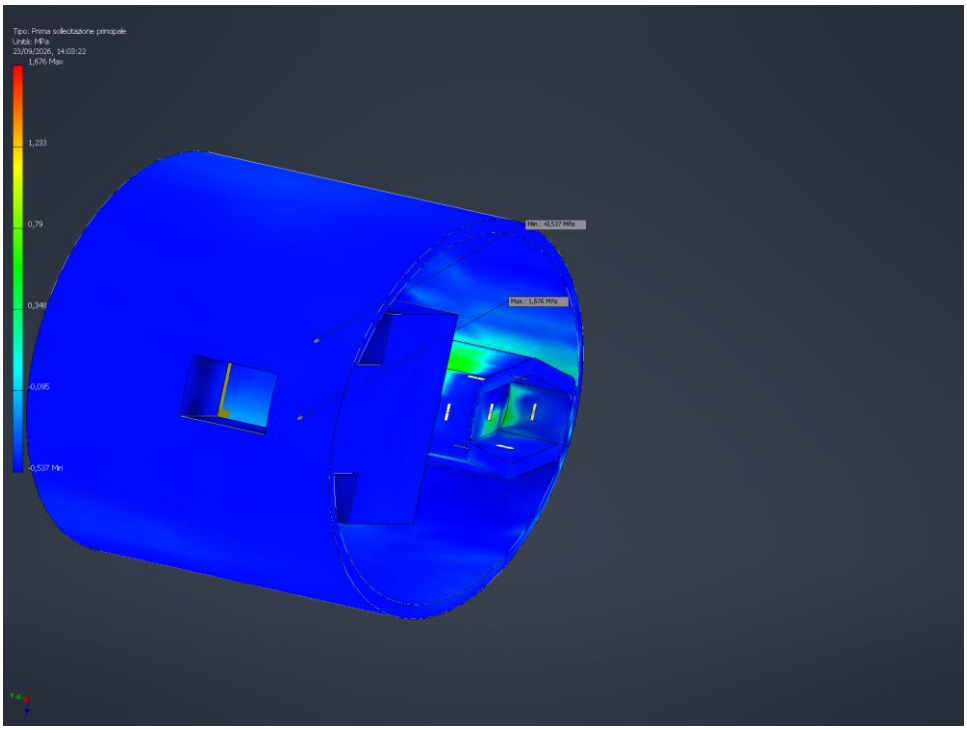
## Figure

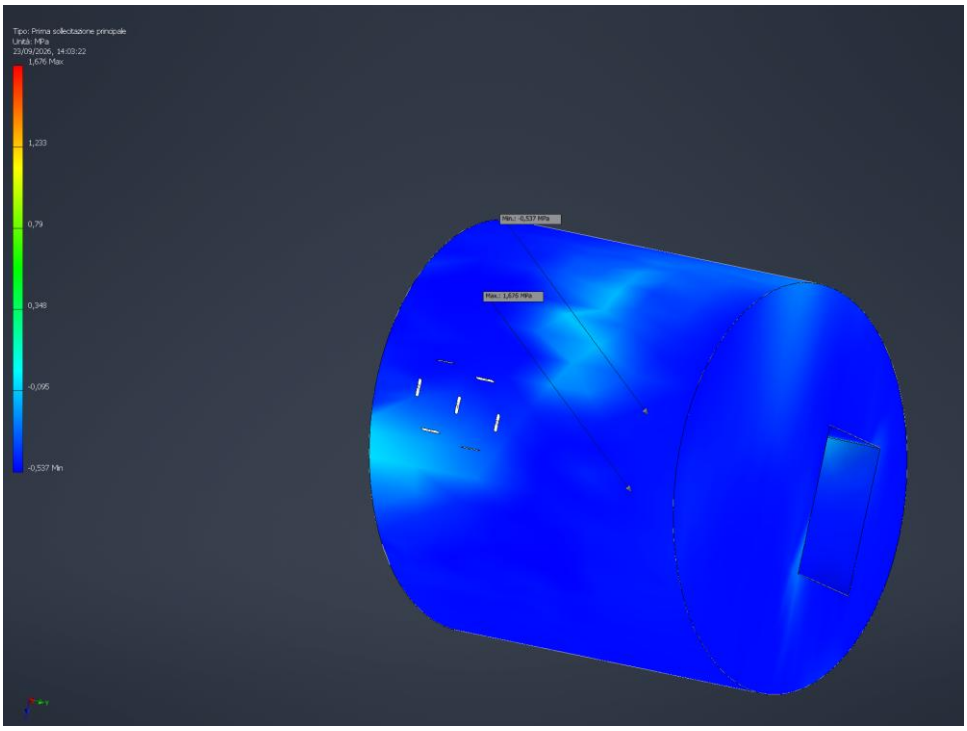
### Sollecitazione di Von Mises



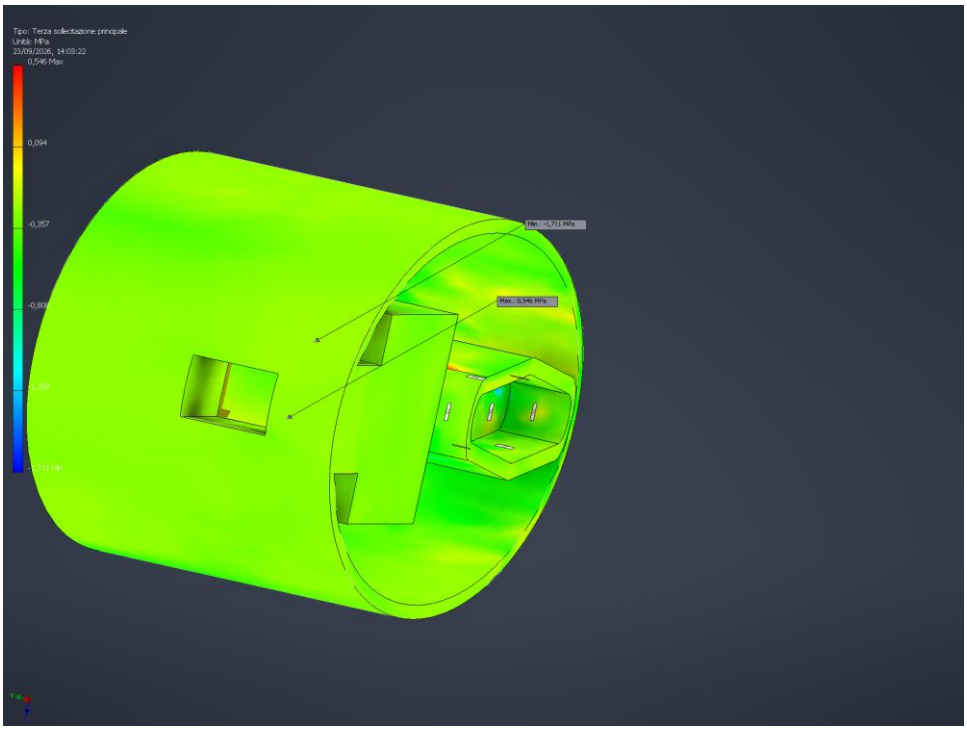



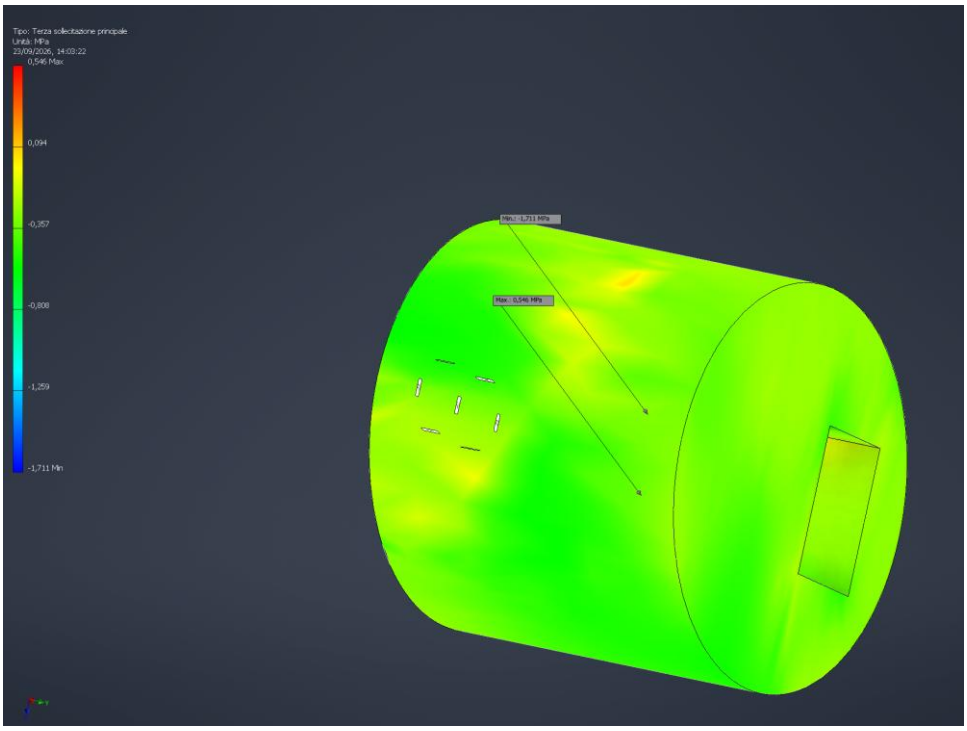

**Prima sollecitazione principale**



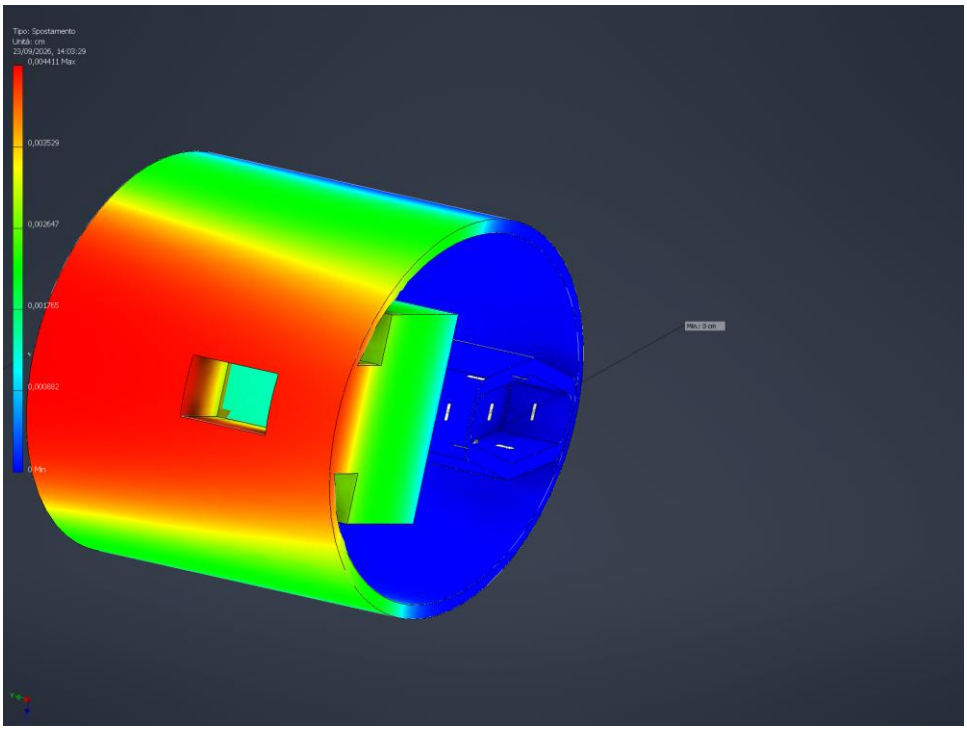



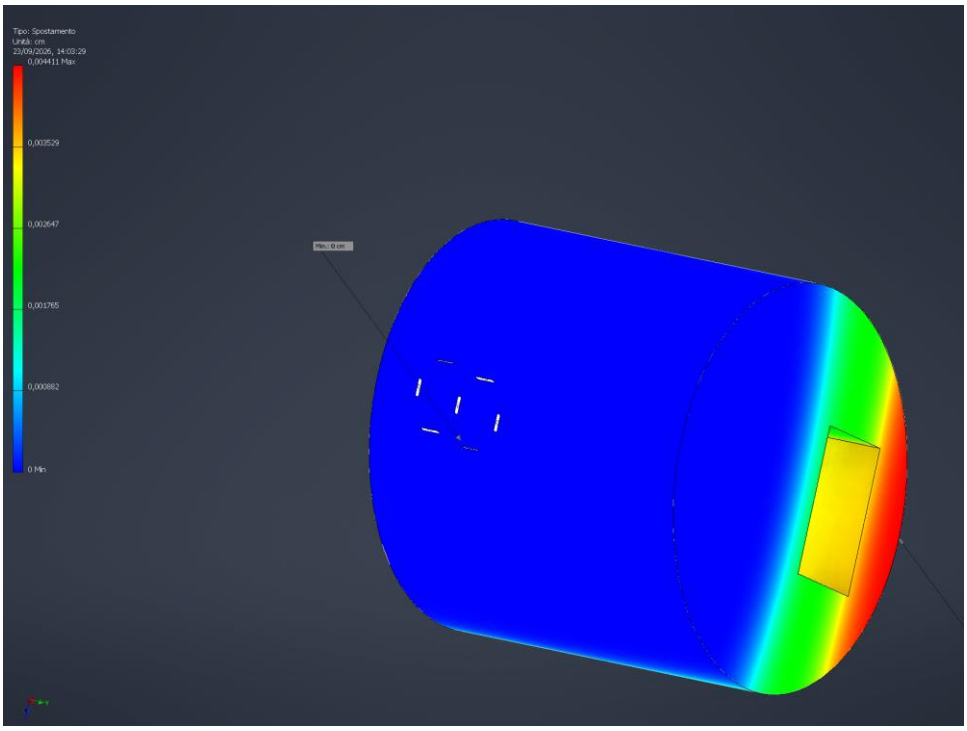

**Terza sollecitazione principale**



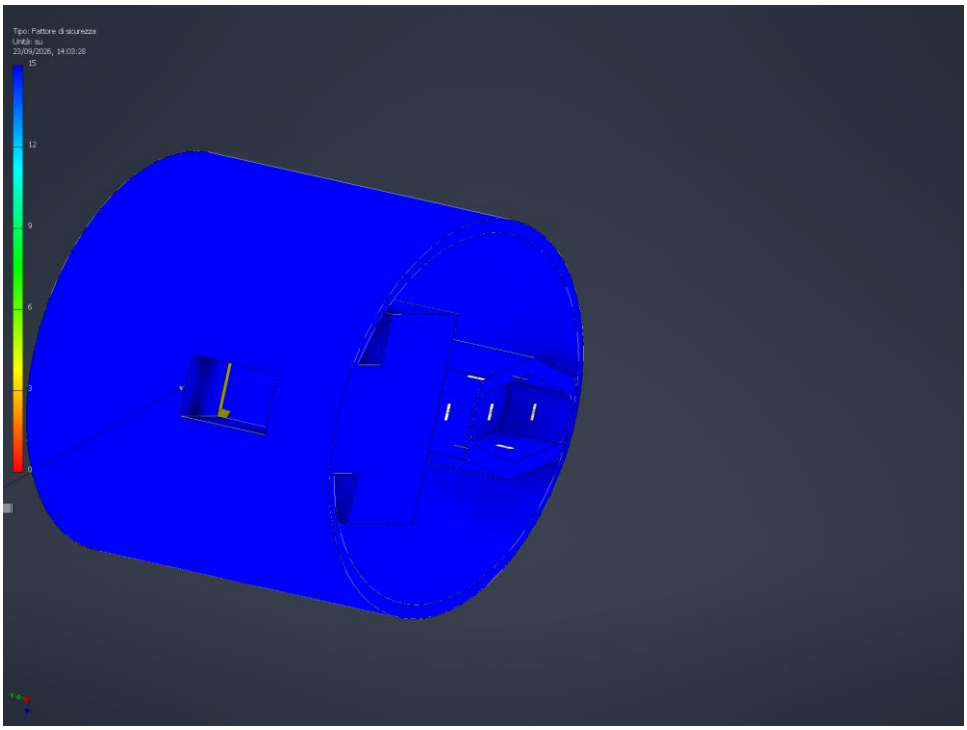



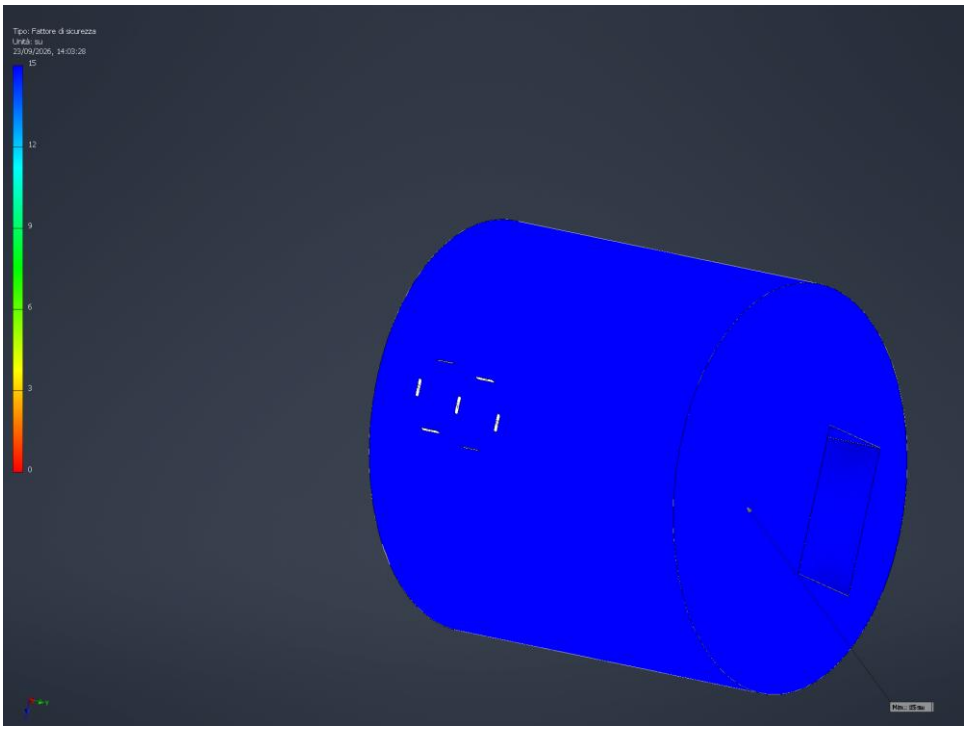

**Spostamento**



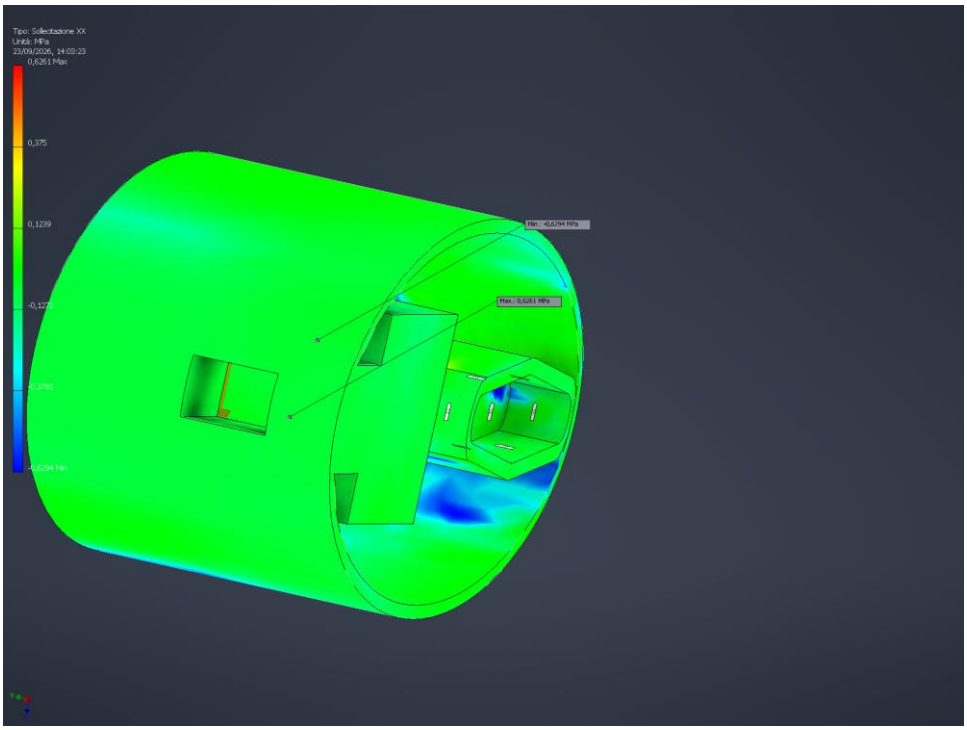



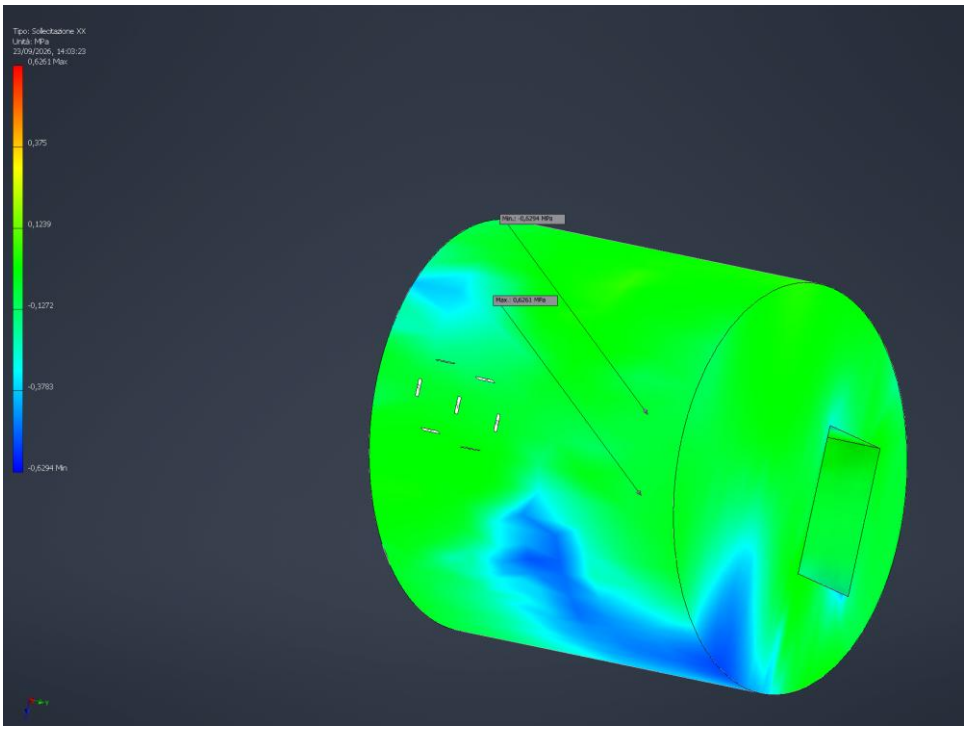

**Fattore di sicurezza**



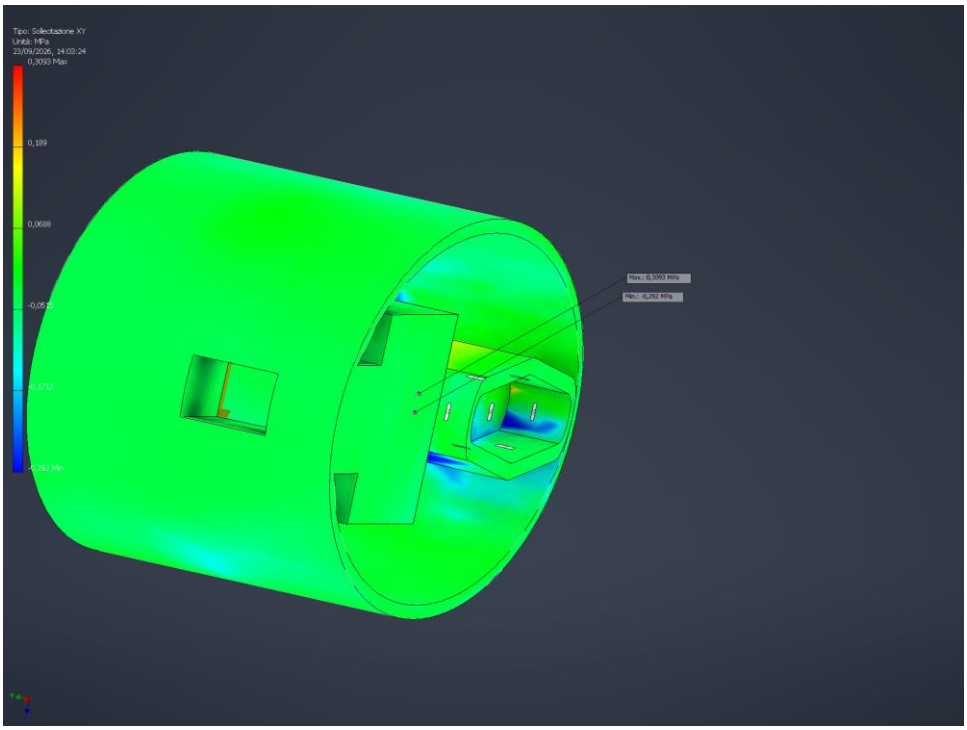



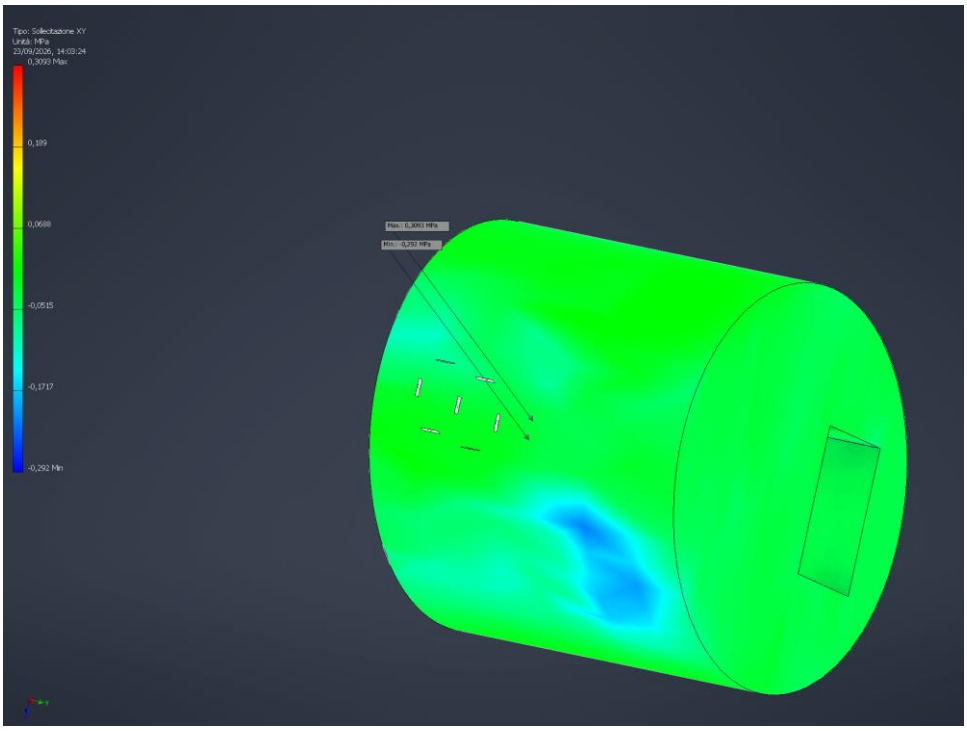

**Sollecitazione XX**



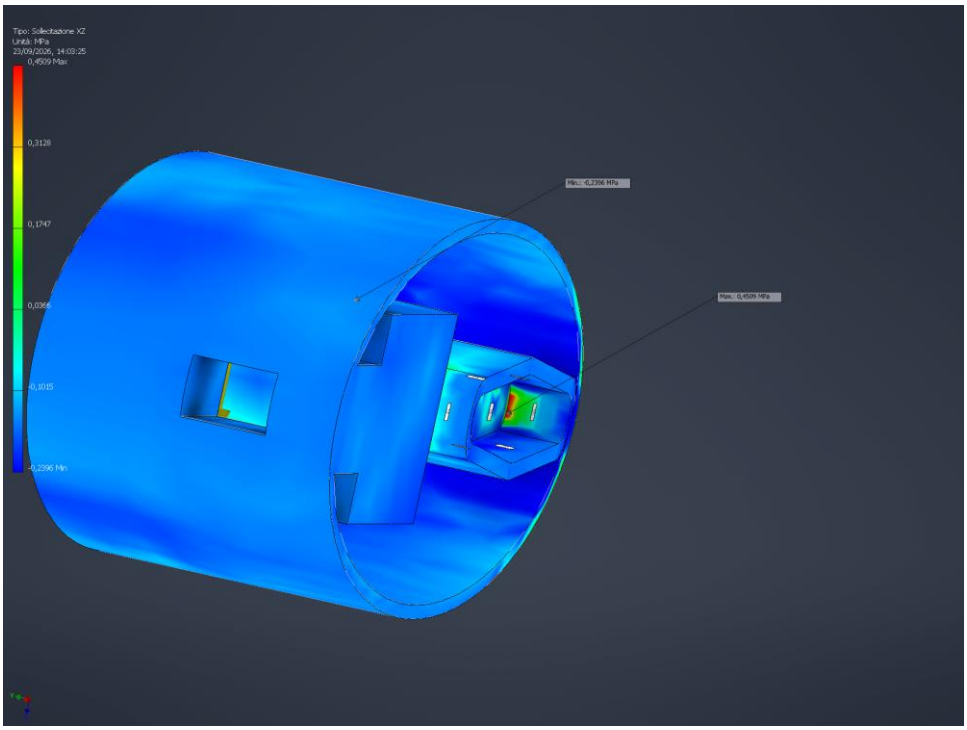



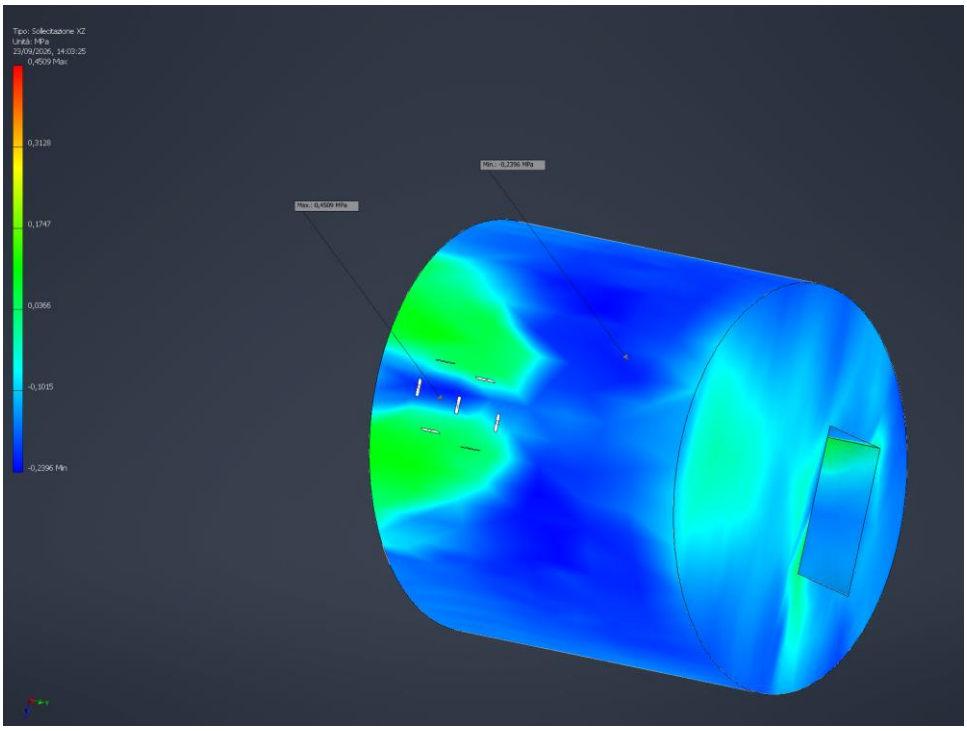

**Sollecitazione XY**



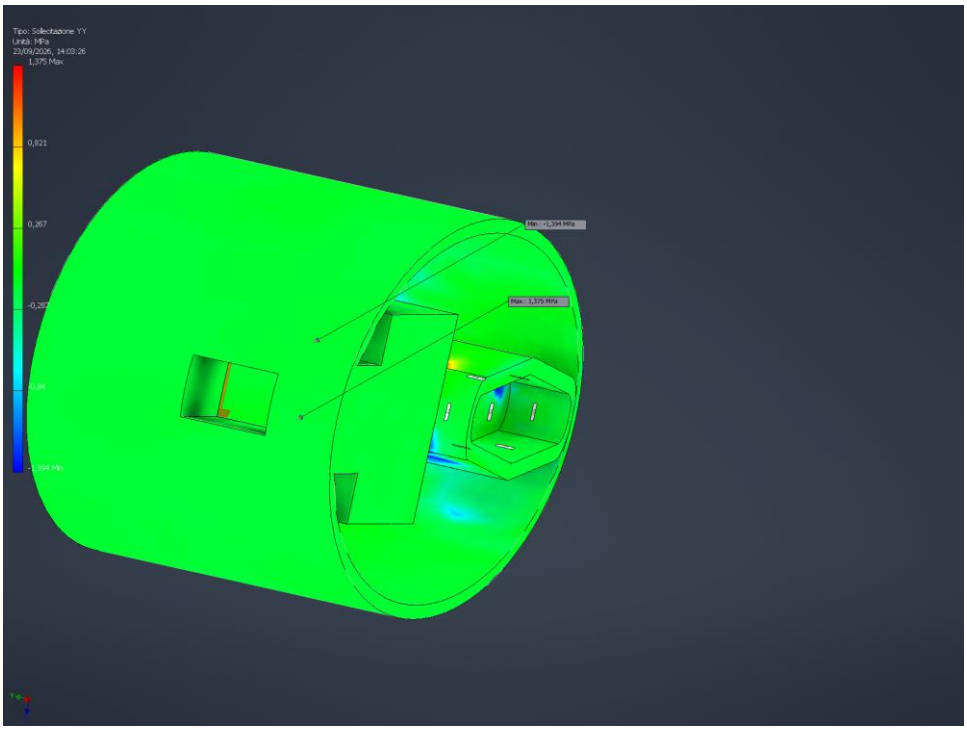



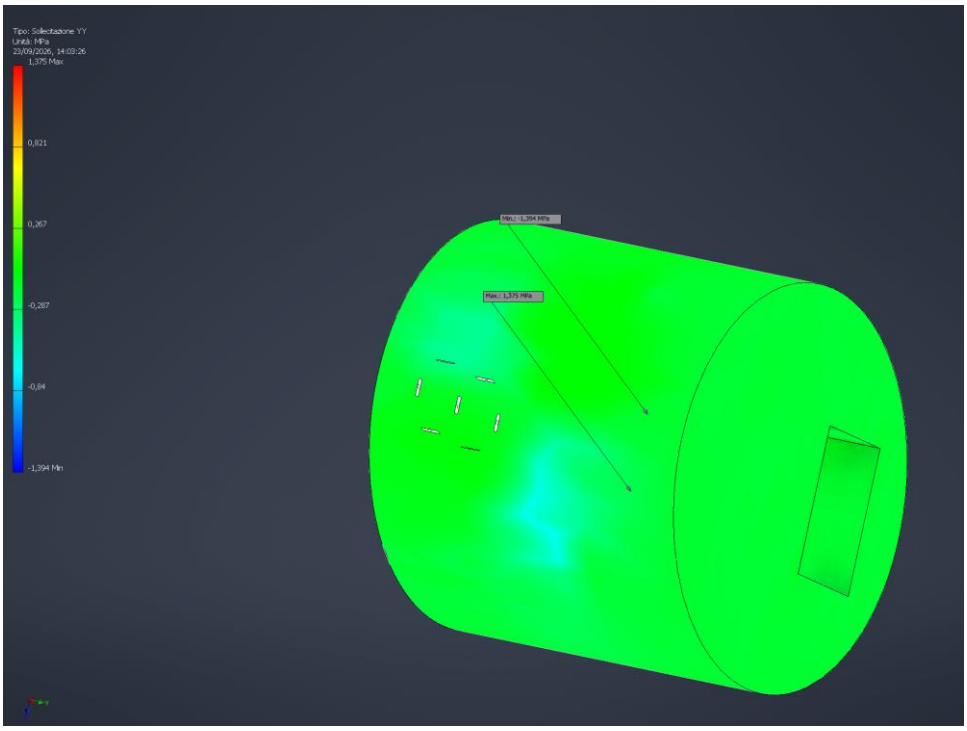

**Sollecitazione XZ**



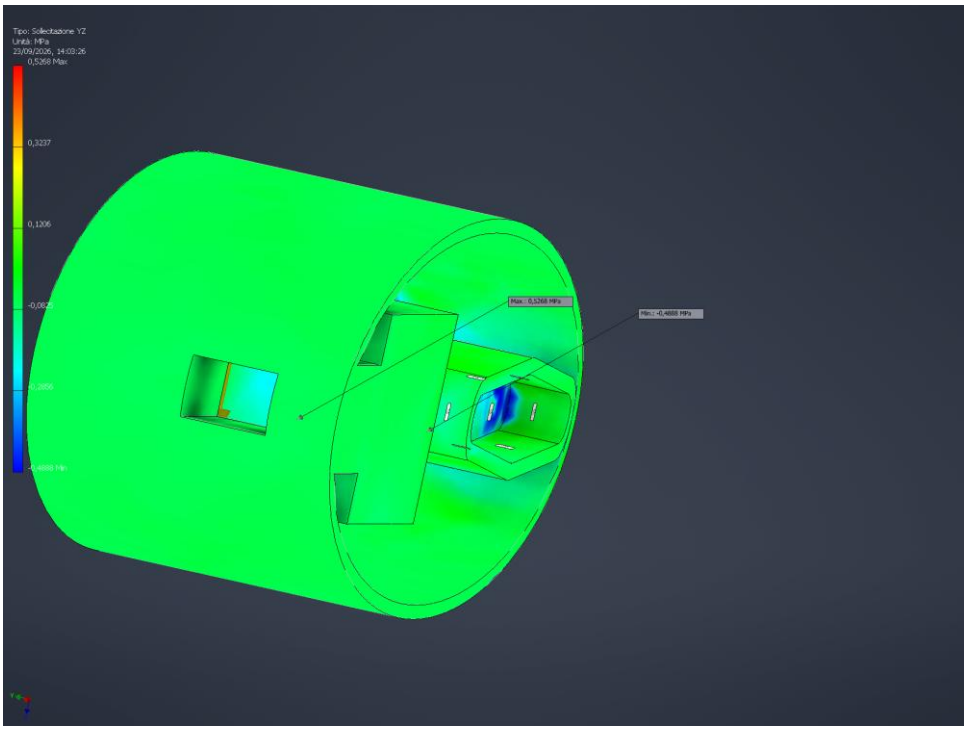



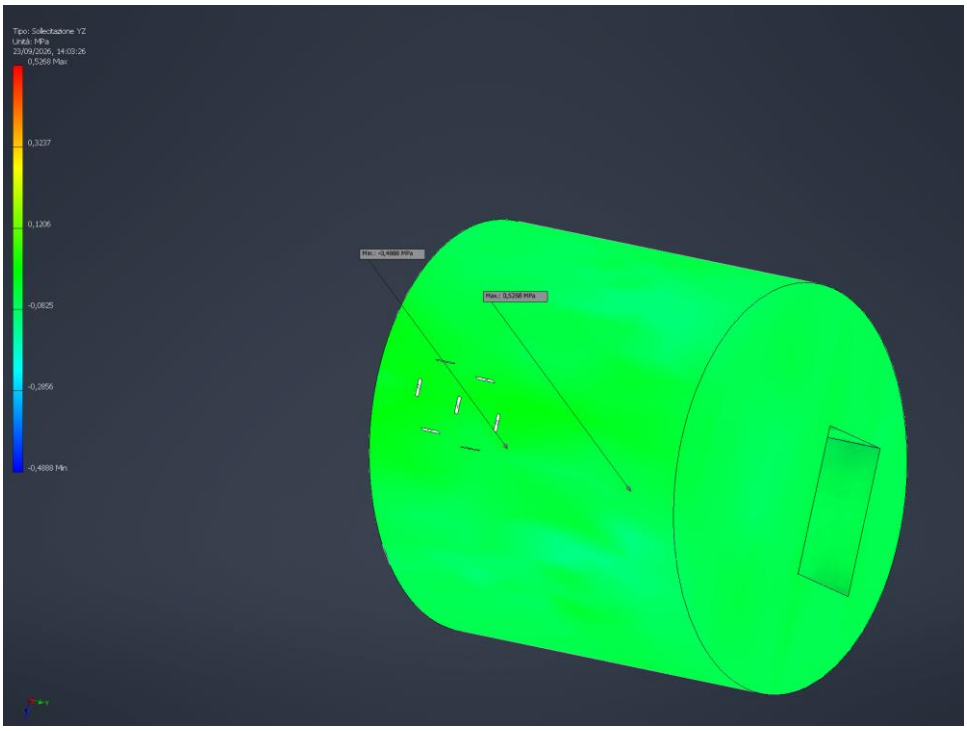

**Sollecitazione YY**



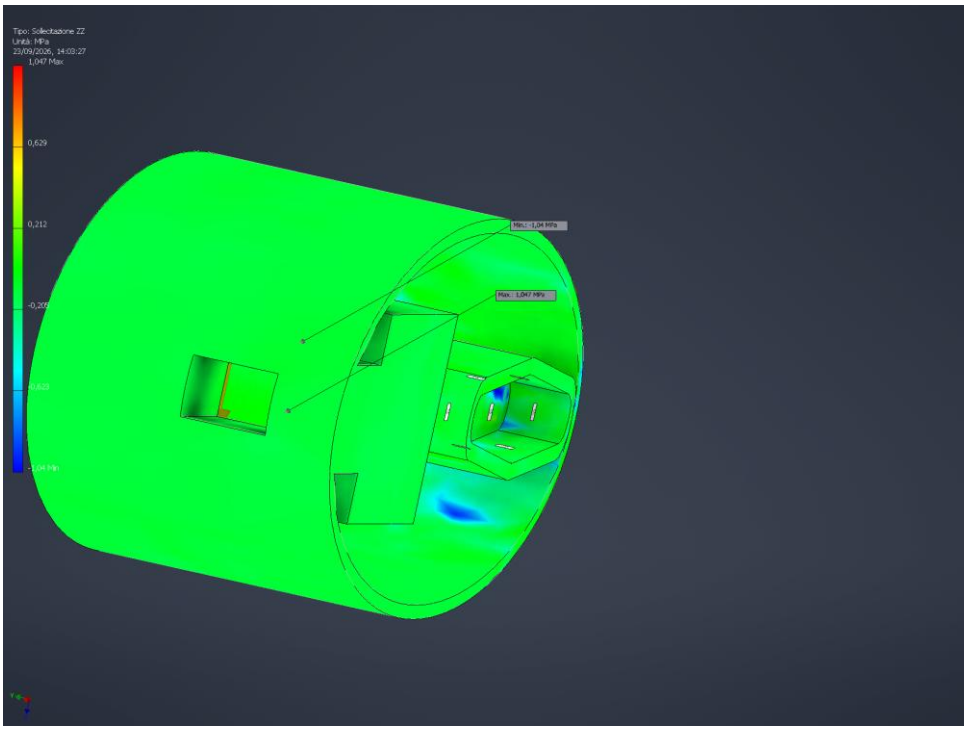



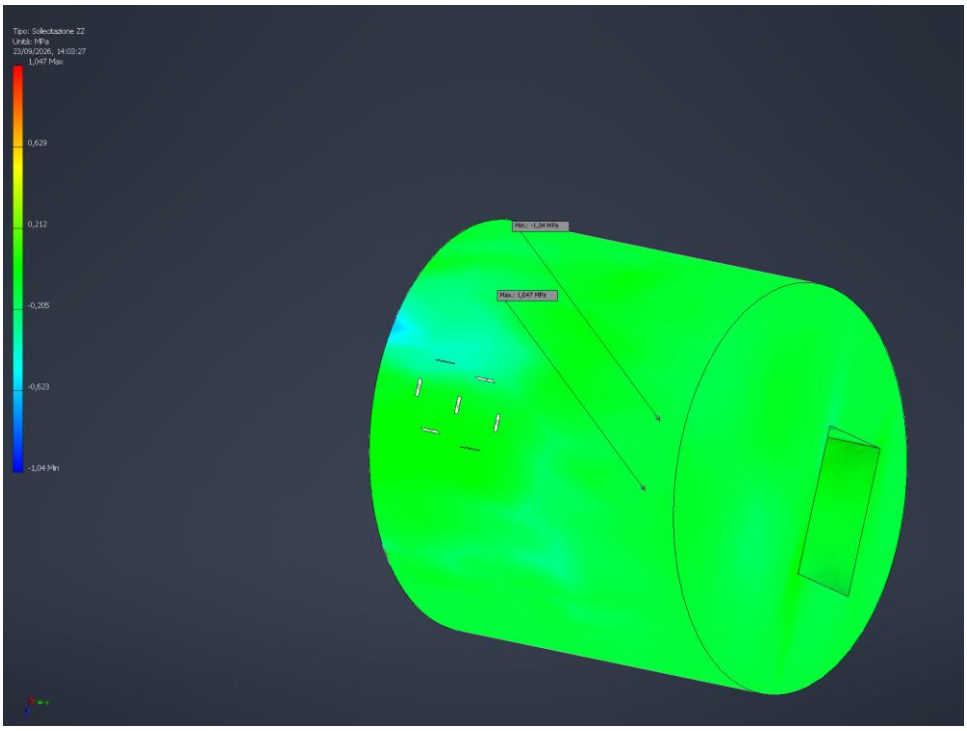

**Sollecitazione YZ**



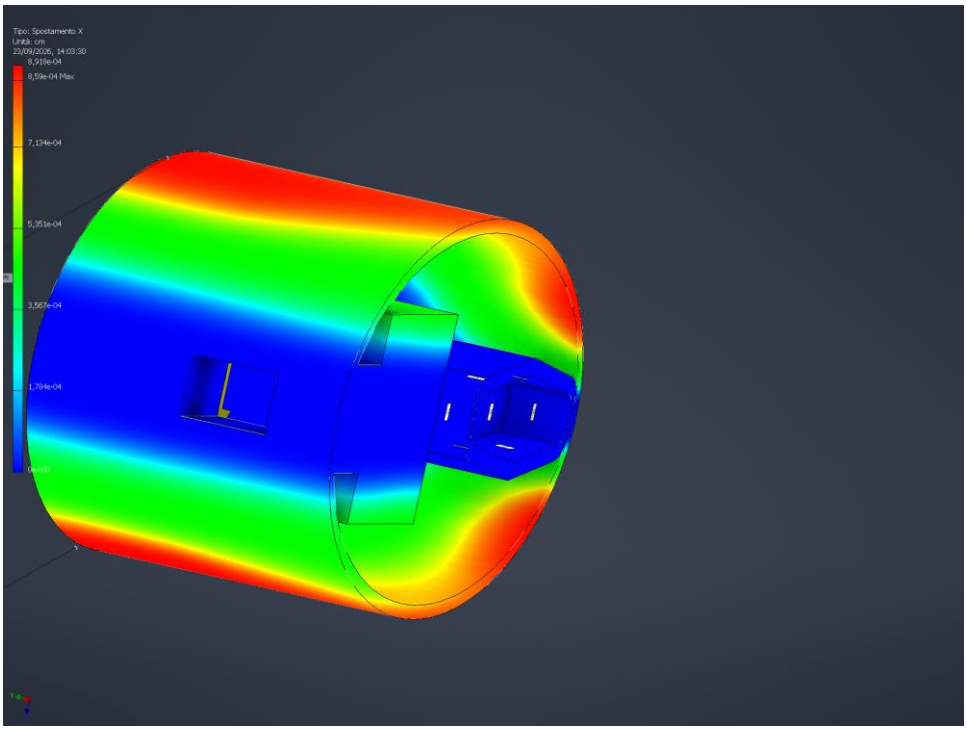



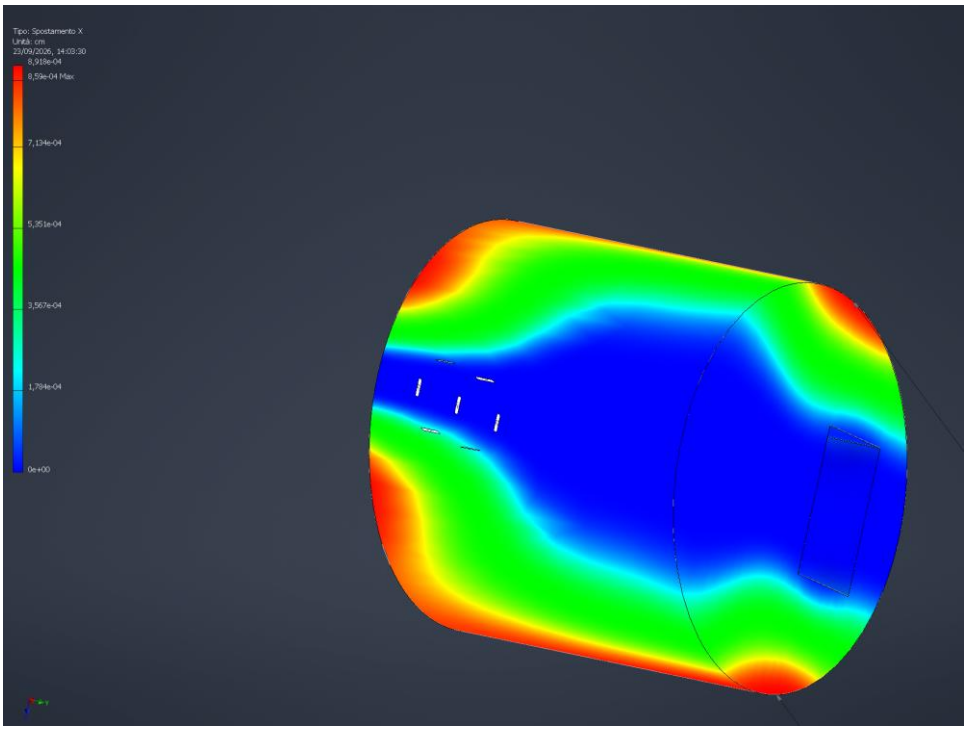

**Sollecitazione ZZ**



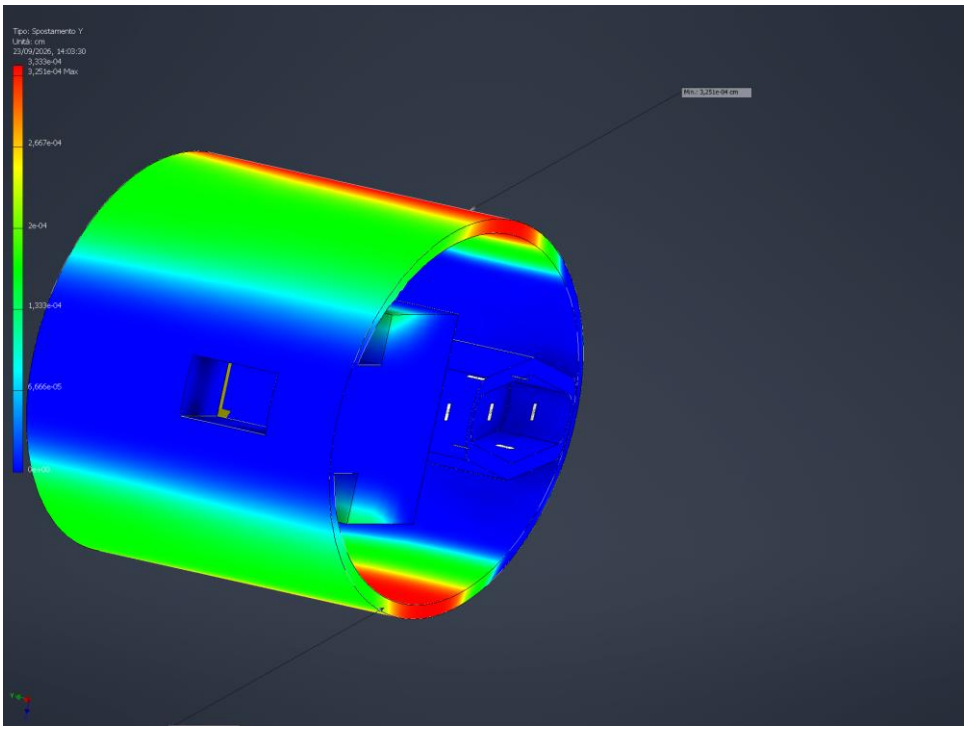



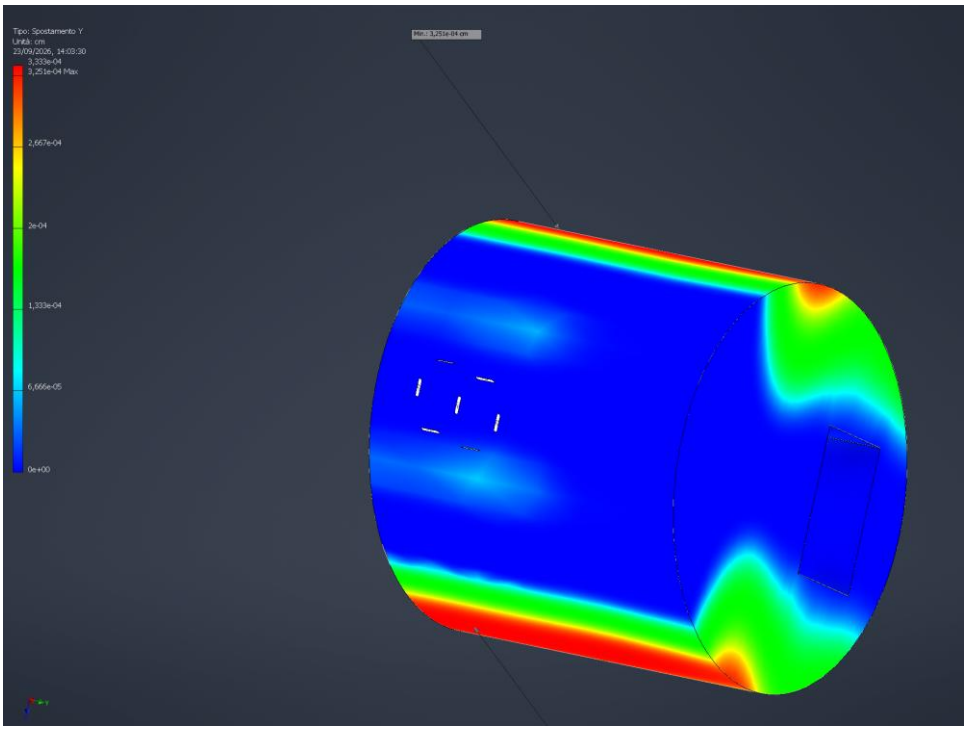

**Spostamento X**



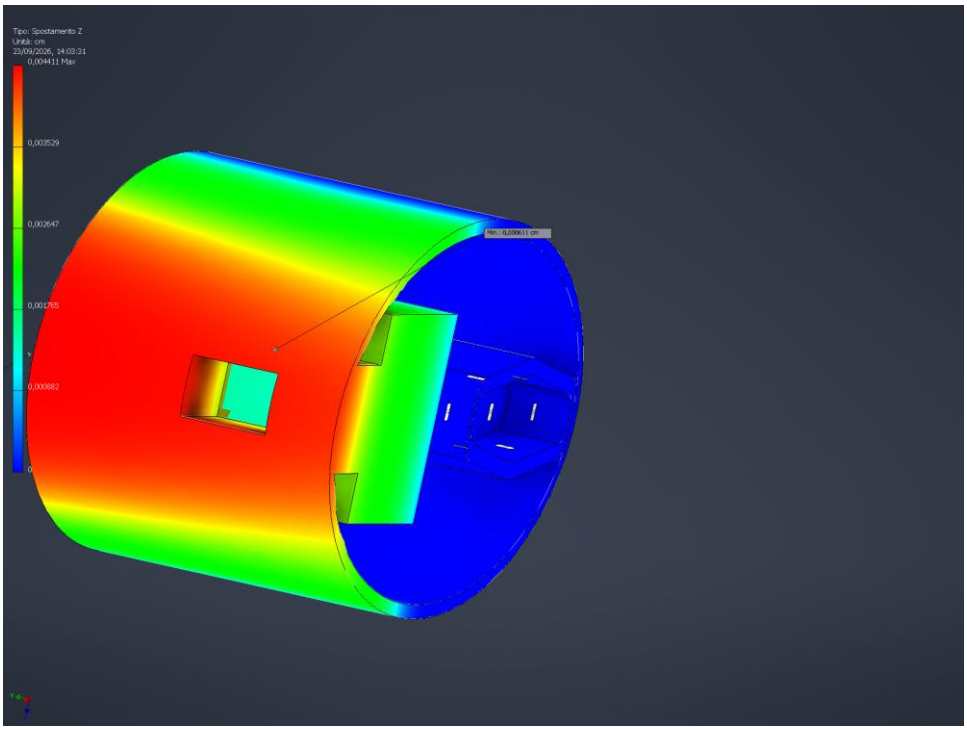



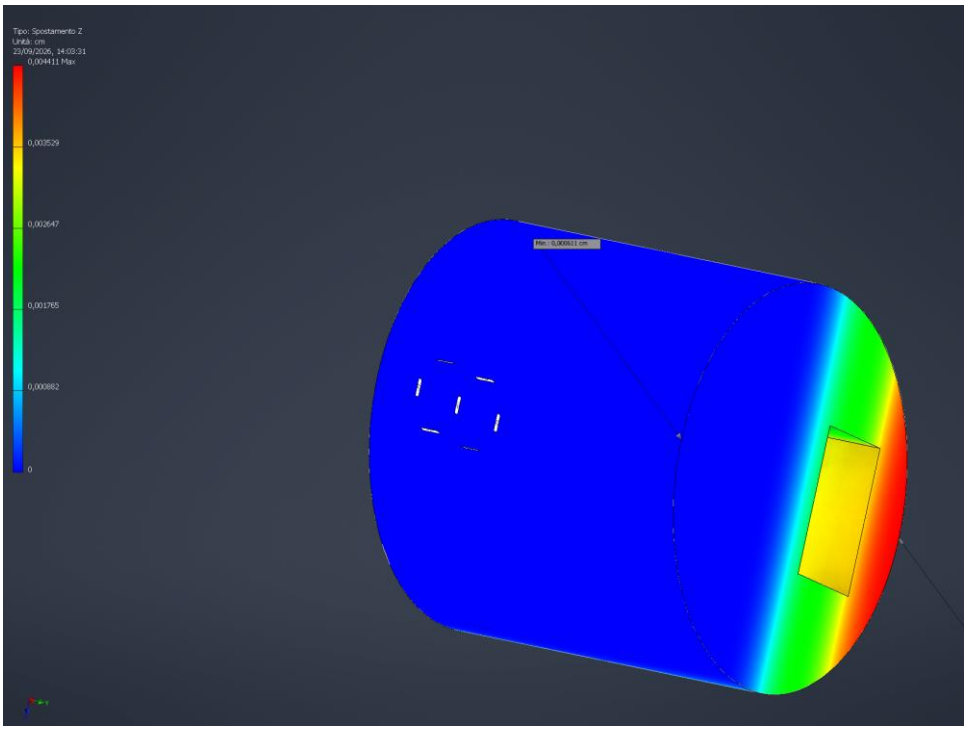

**Spostamento Y**



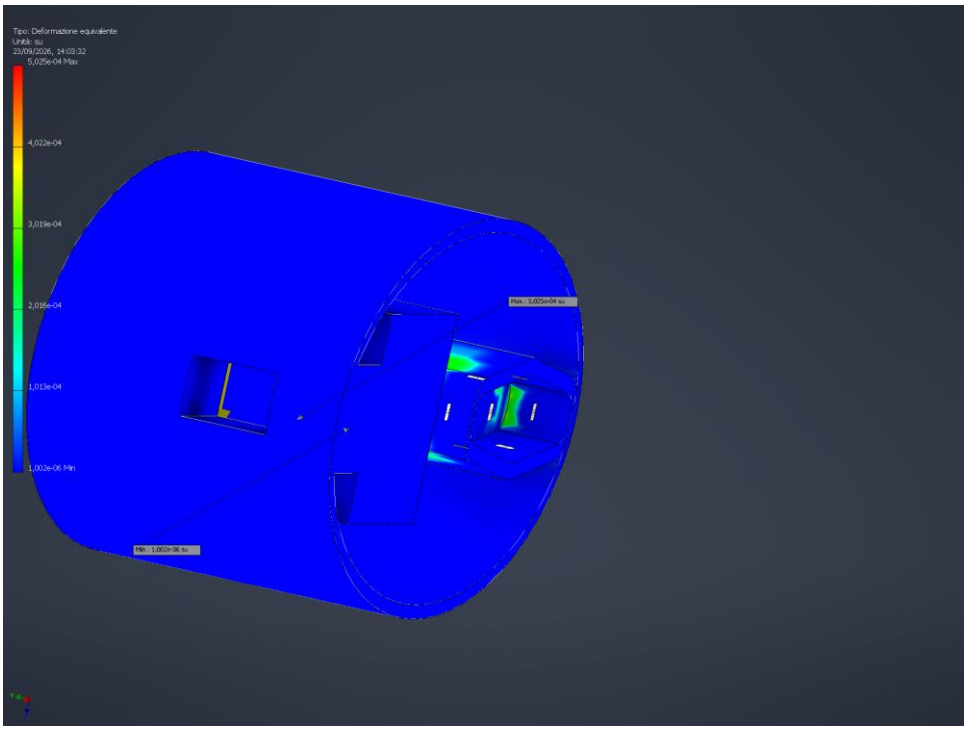



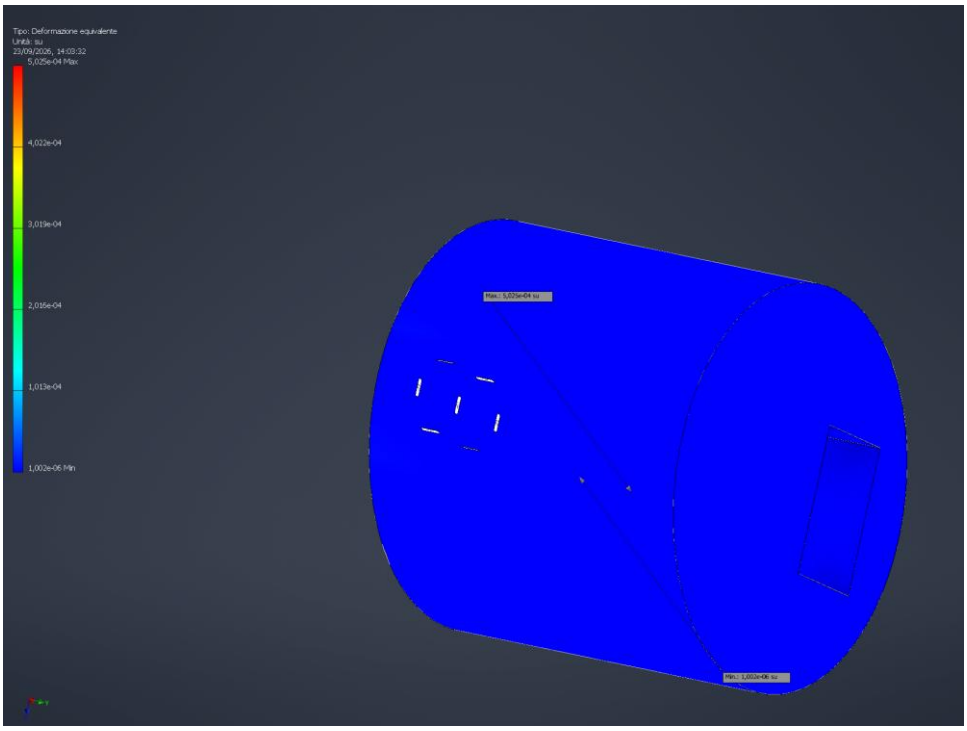

**Spostamento Z**



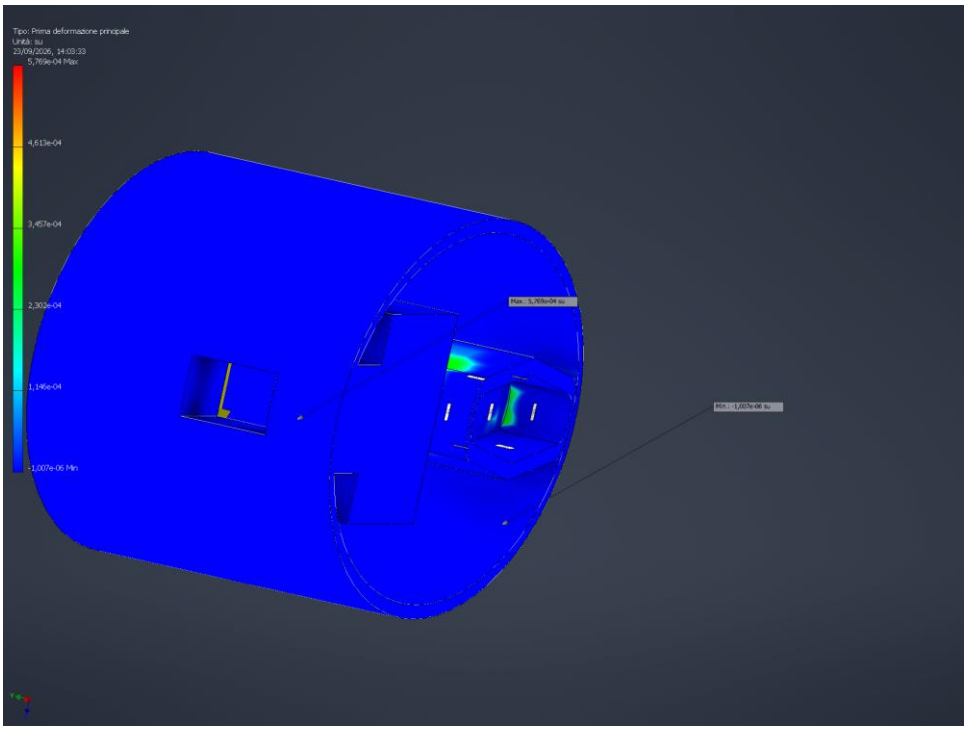



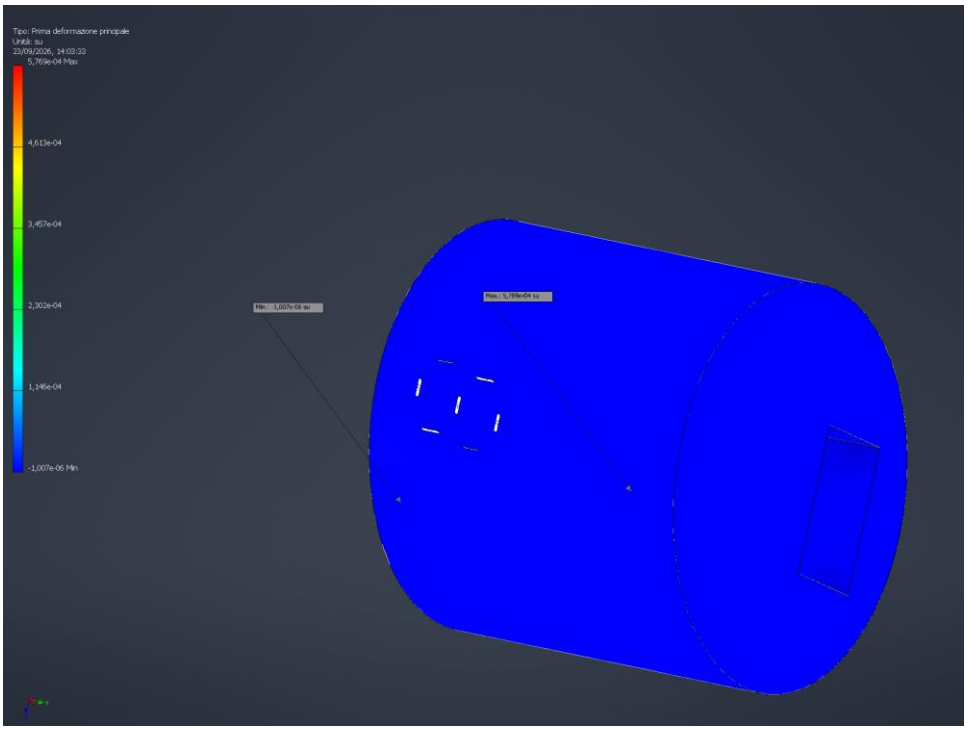

**Deformazione equivalente**



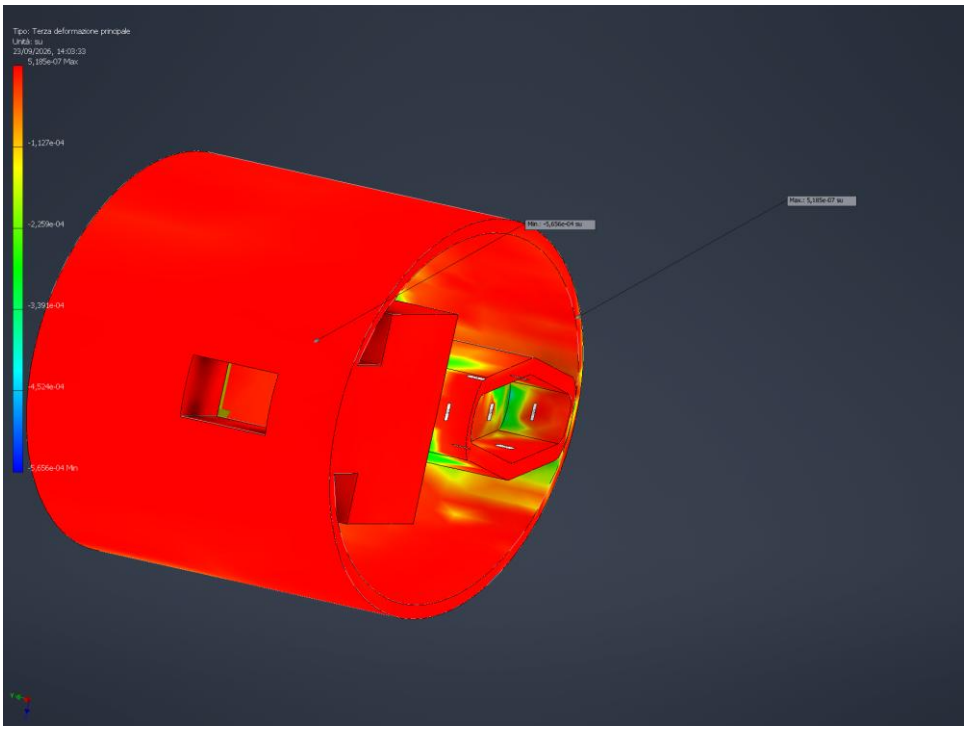



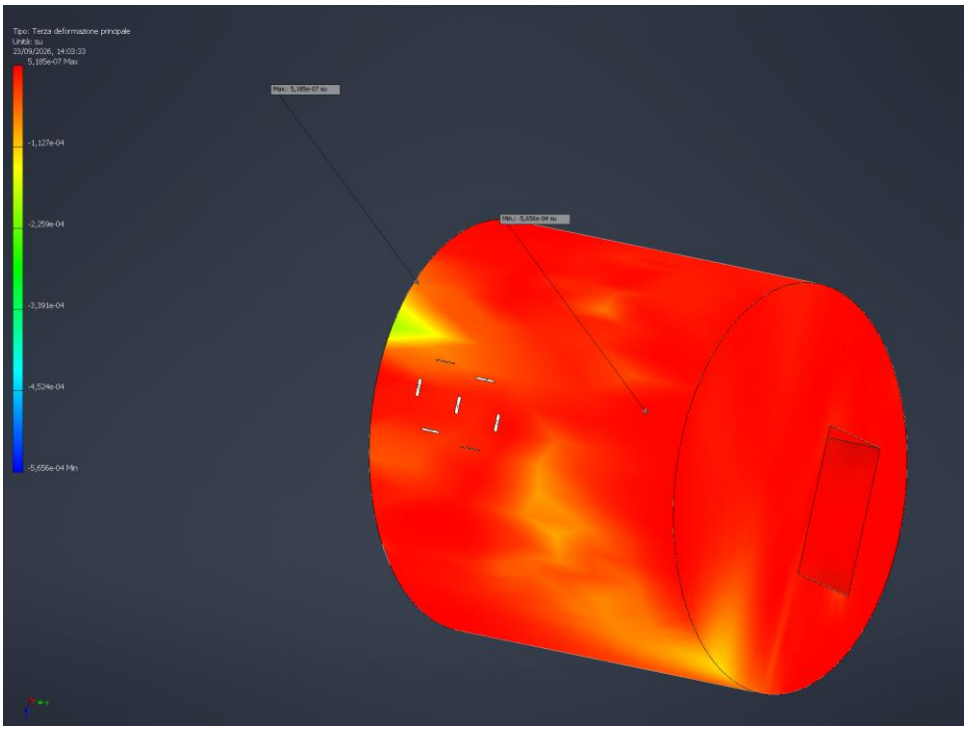

**Prima deformazione principale**



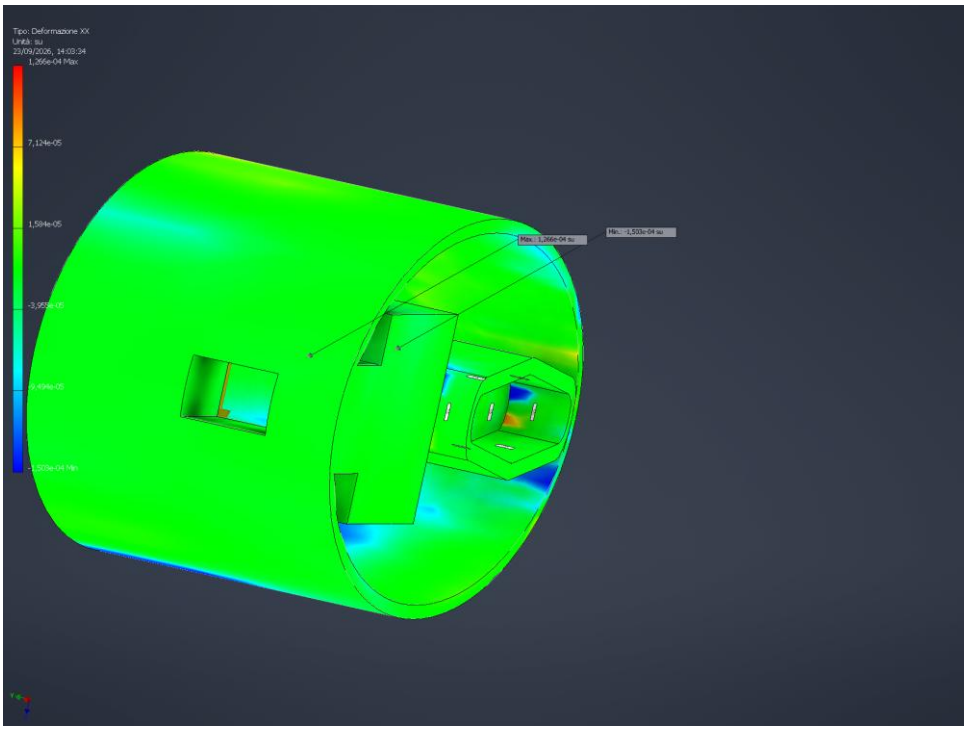



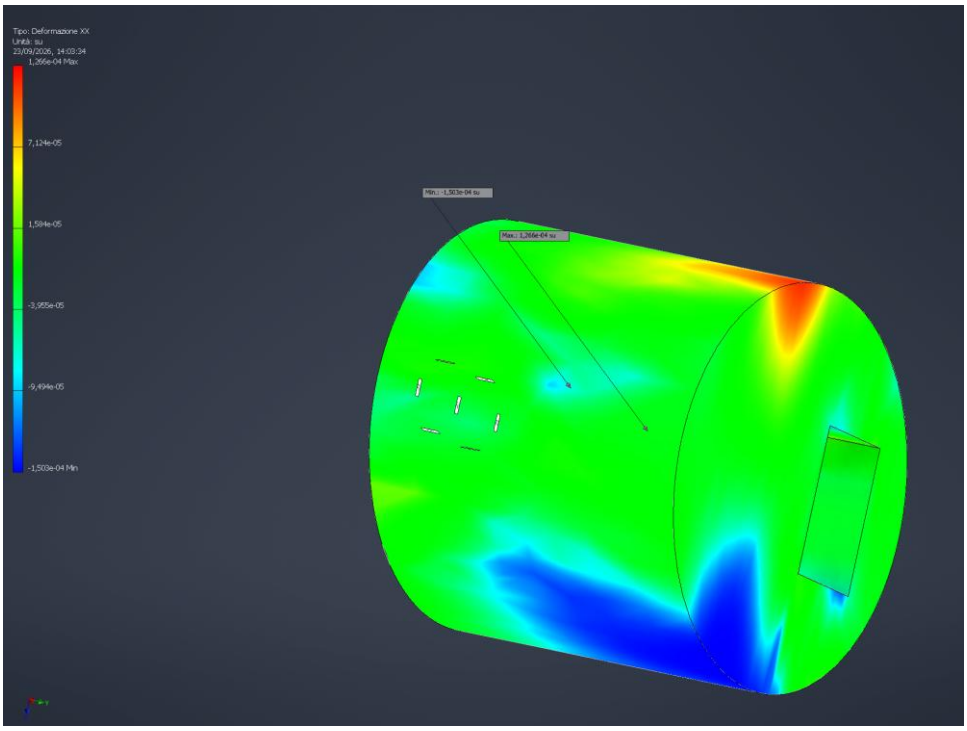

**Terza deformazione principale**



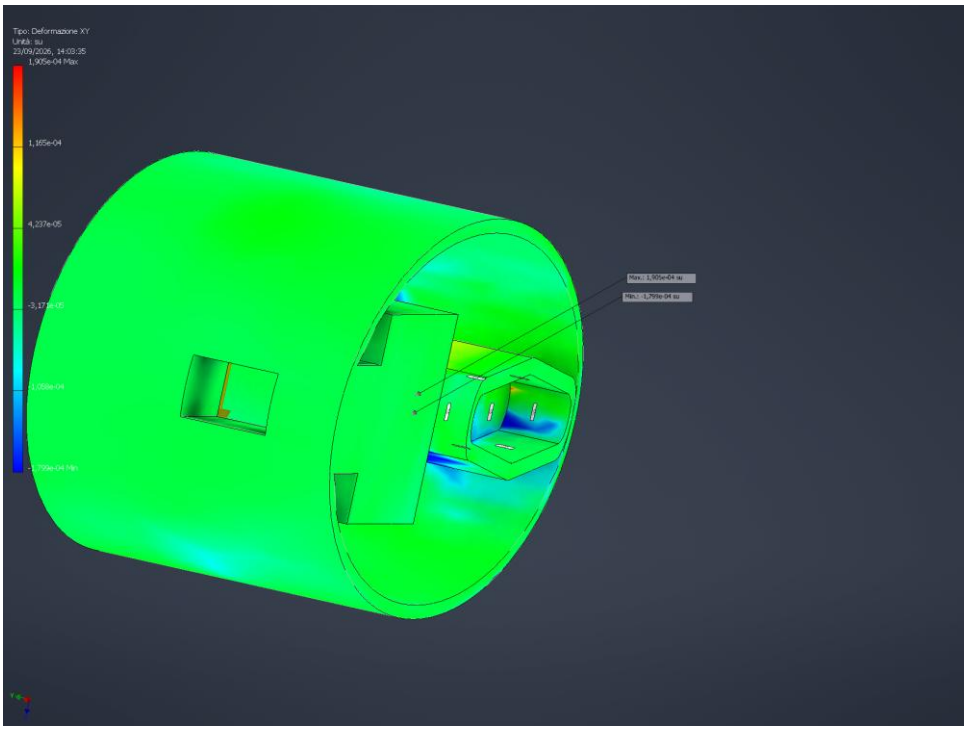



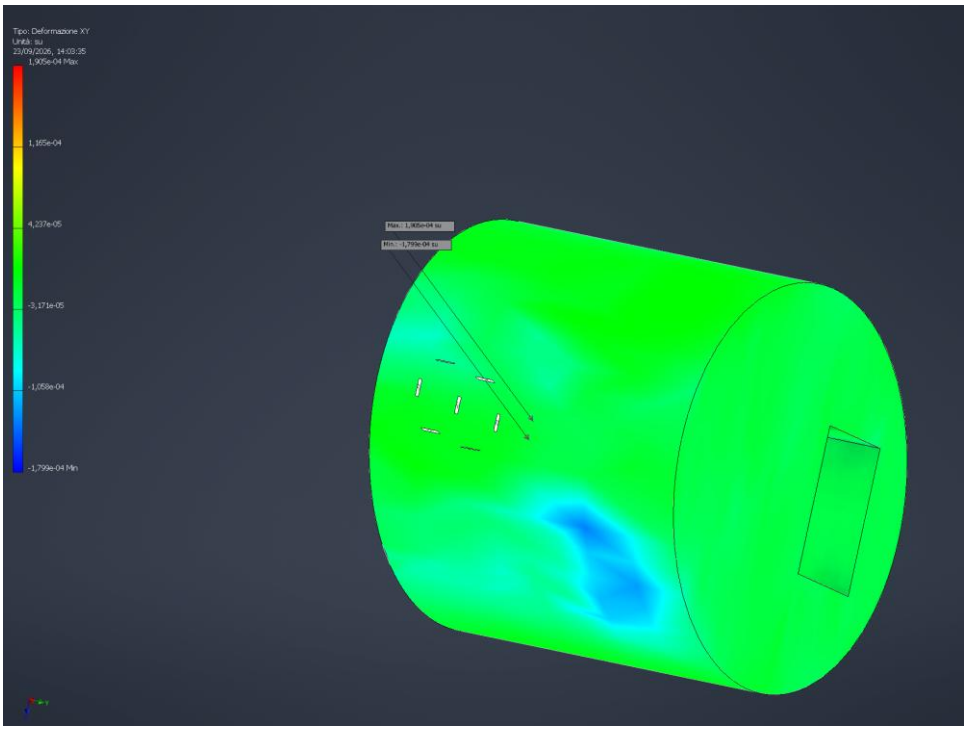

**Deformazione XX**



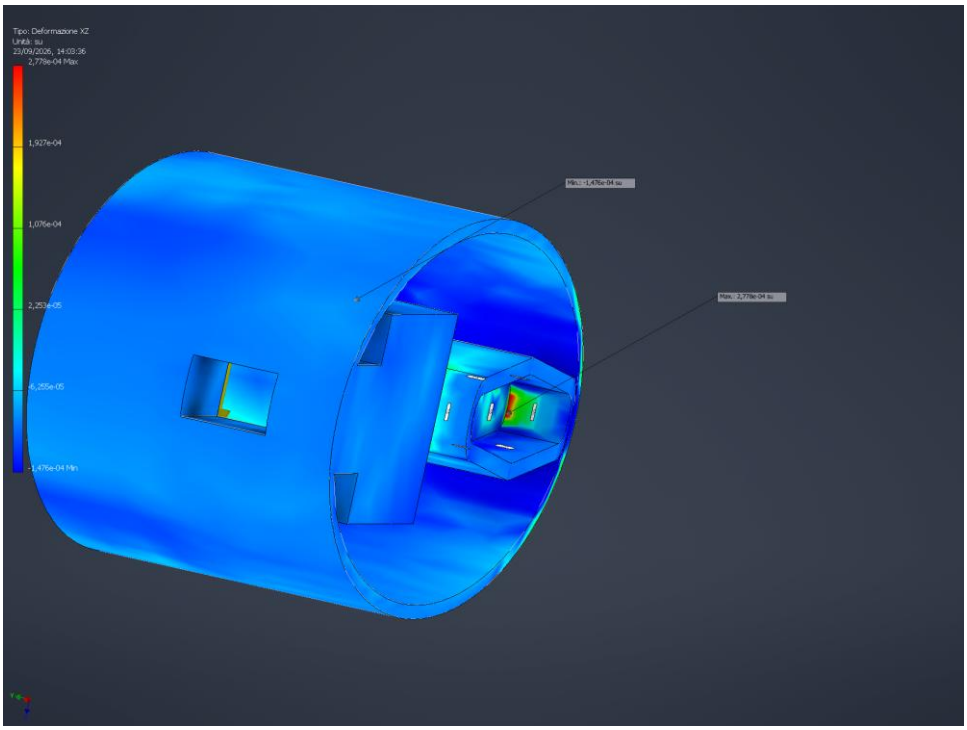



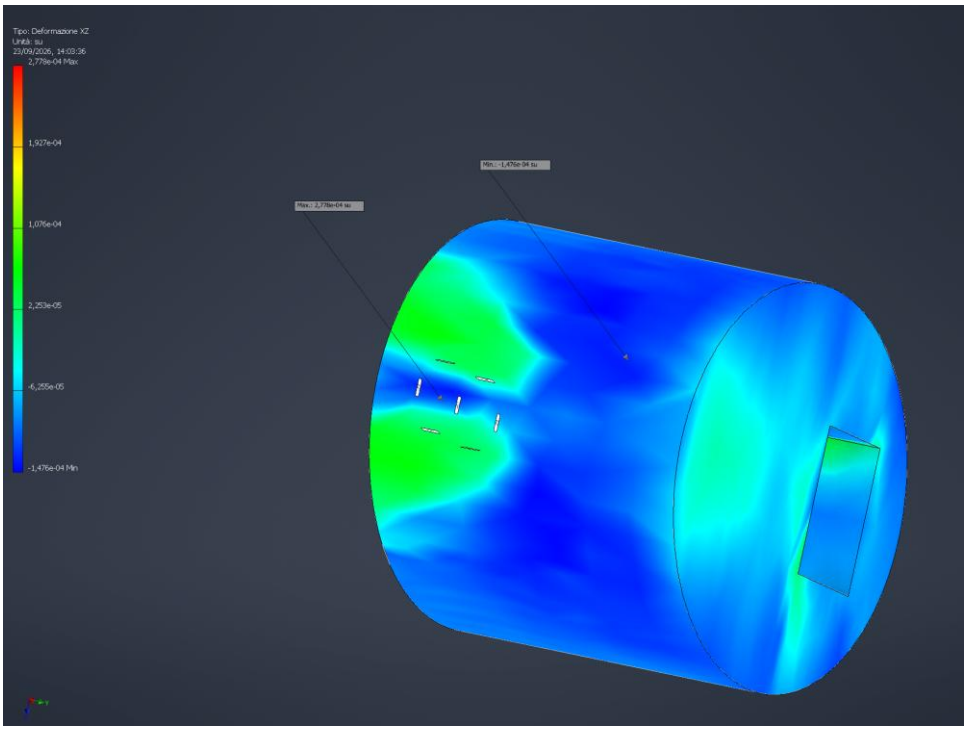

**Deformazione XY**



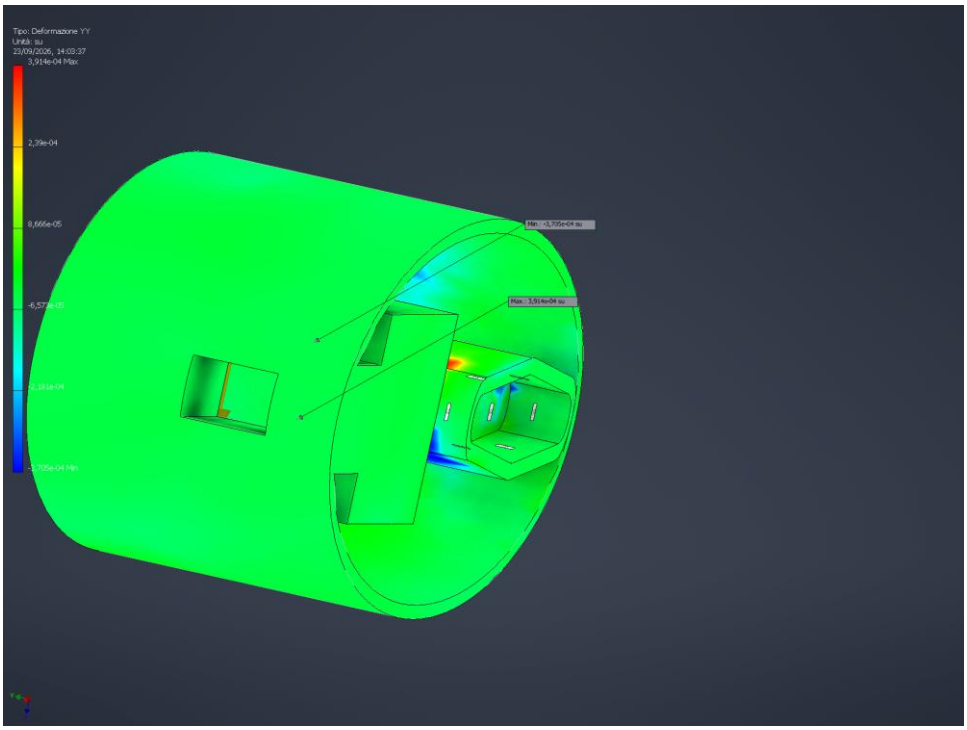



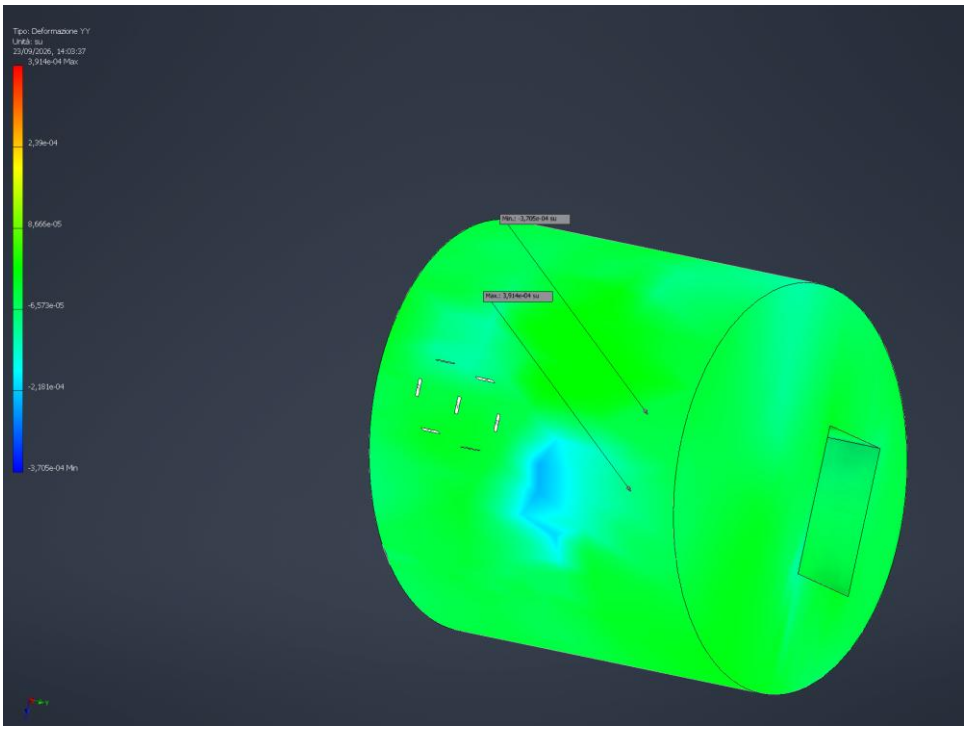

**Deformazione XZ**



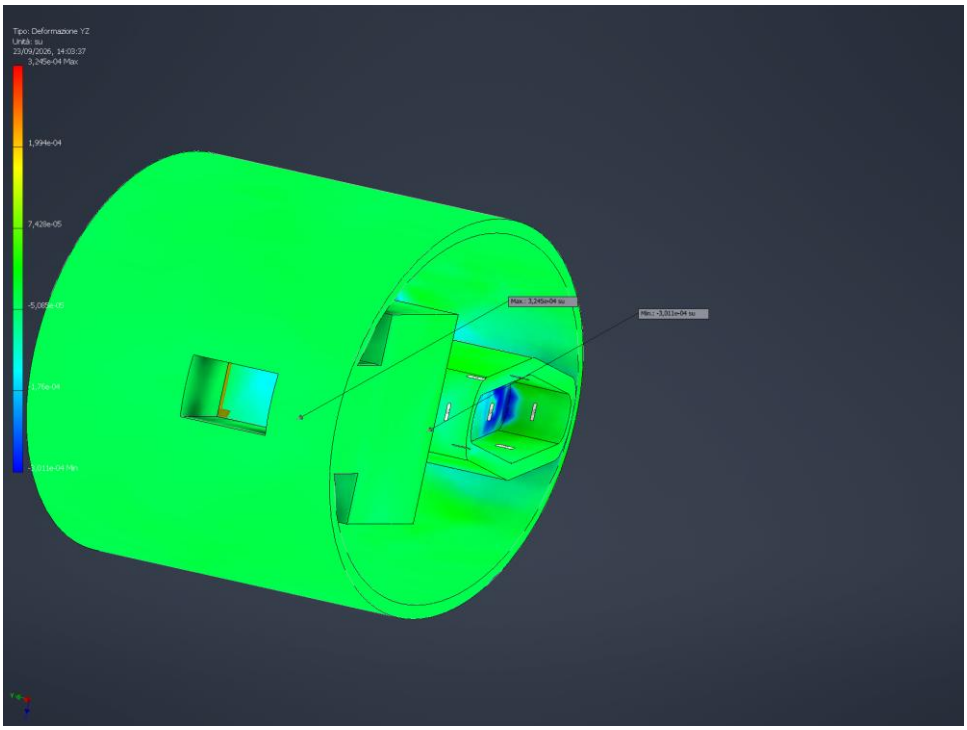



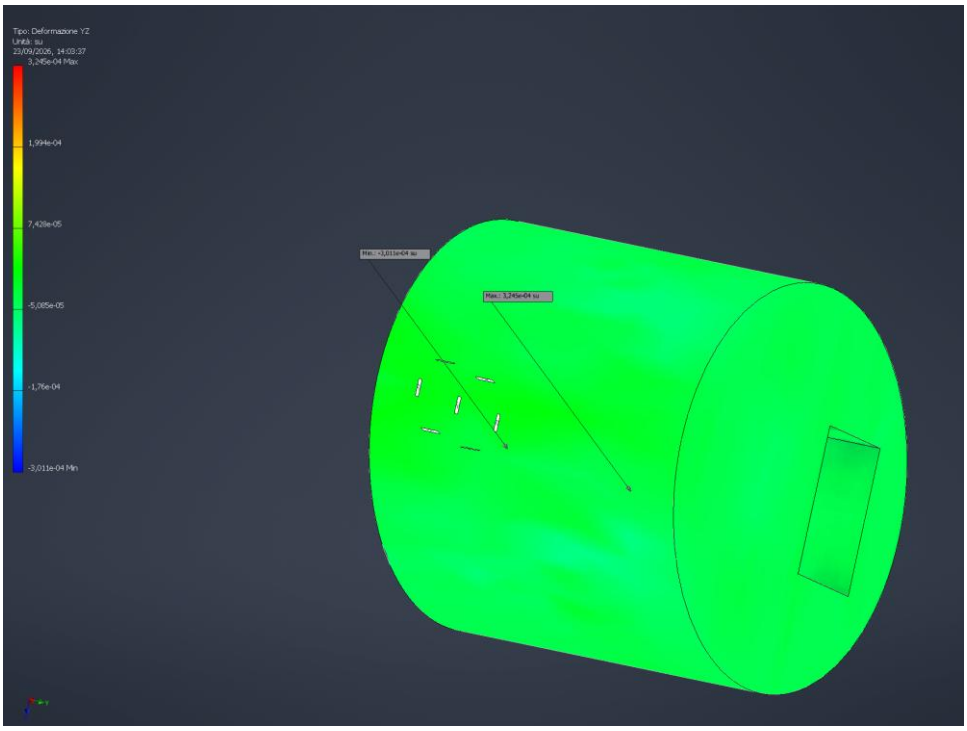

**Deformazione YY**



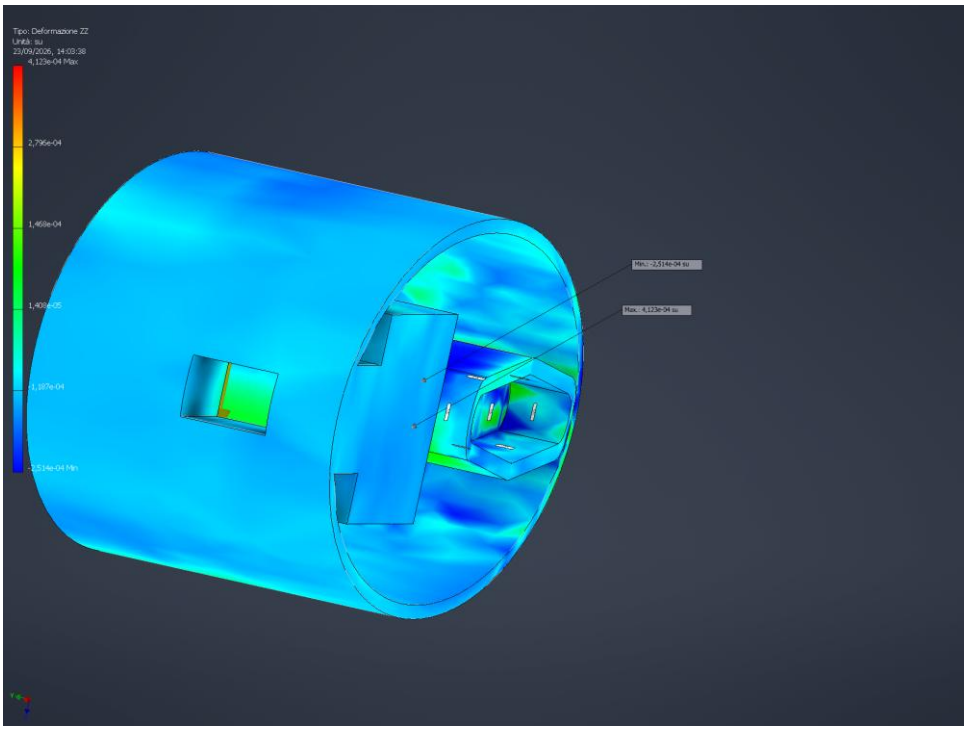



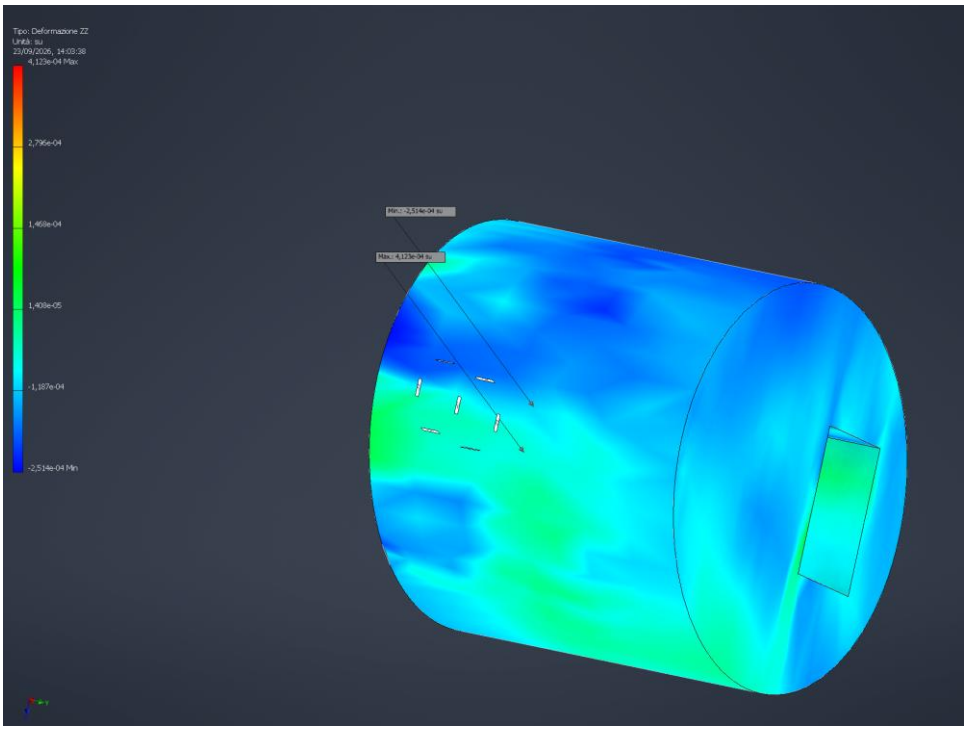

**Deformazione YZ**




**Deformazione ZZ**




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